WindowsCE.NET BSP Build Guide

MMSP™2

User Manual

Ver. 2.4.2 beta / July 27, 2005

MP2520F

User's manual for MP2520F DTK 4.0 with Microsoft Windows® CE.NET 4.2

NOTICE

MAGICEYES INFORMS THAT THIS CODE AND INFORMATION IS PROVIDED "AS IS" BASE AND WITHOUT WARRANTY OF ANY KIND, EITHER EXPRESSED OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE IMPLIED WARRANTIES OF MERCHANTABILITY AND/OR FITNESS FOR A PARTICULAR PURPOSE.

I N D E X

WinCE.NET BSP Build Guide for MP2520F DTK4.0

1. Creating a New Platform

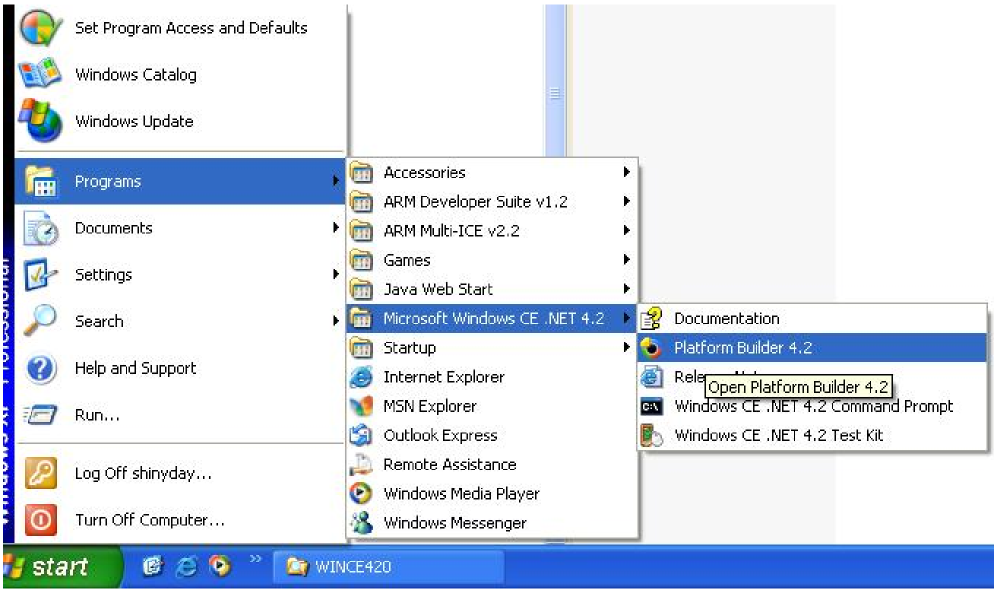
1.1 Installing BSP

Create **MMSP20** folder under **WINCE420\PLATFORM** folder

Uncompress BSP File in CD to **WINCE420\PLATFORM\MMSP20** Folder

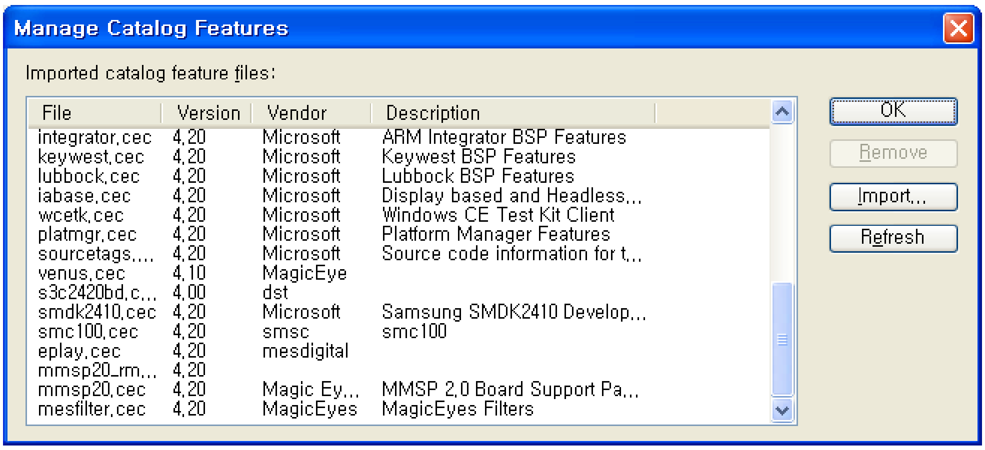
1.2 Catalog Registration of BSP

1. Run Platform Builder

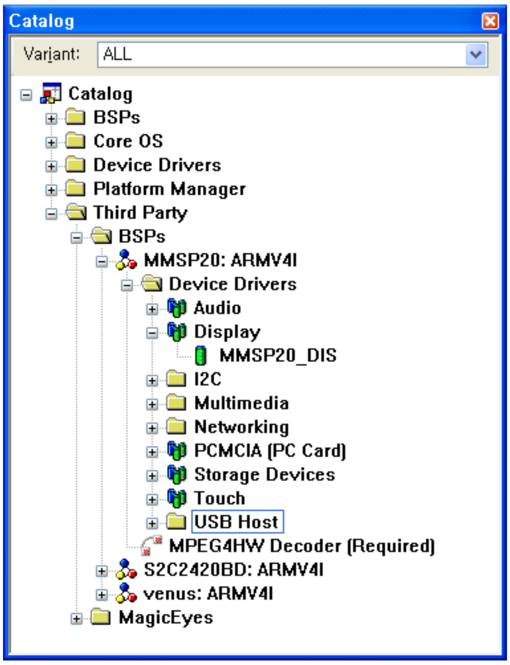


2. Select **Manage Catalog Features** in **File Menu**..

3. Press **Import** button and select **MMSP20.cec** in MMPS20 folder.

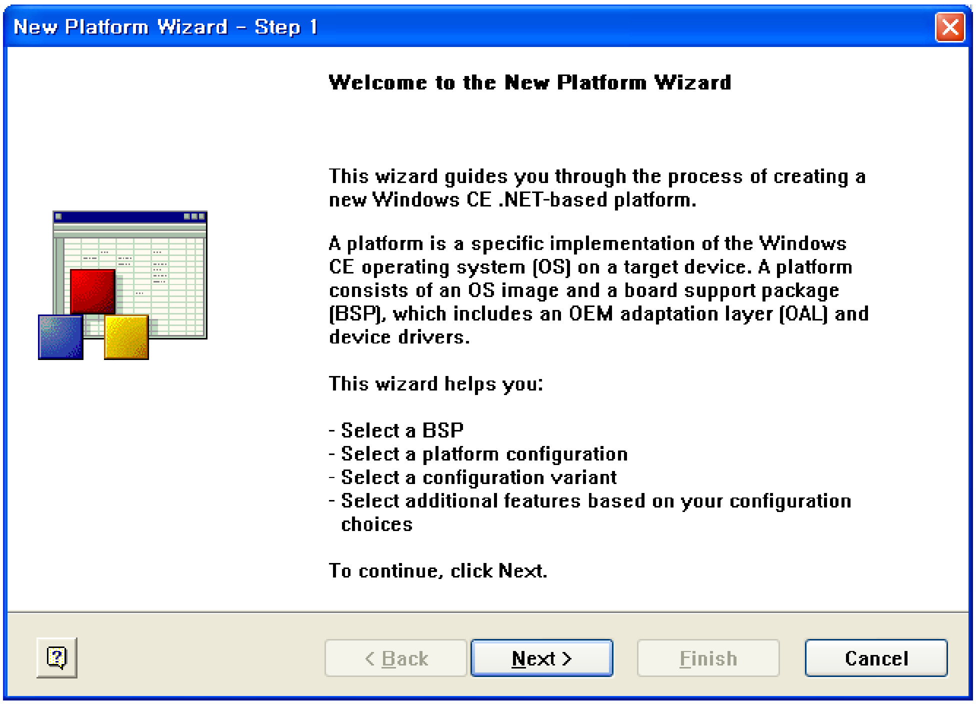


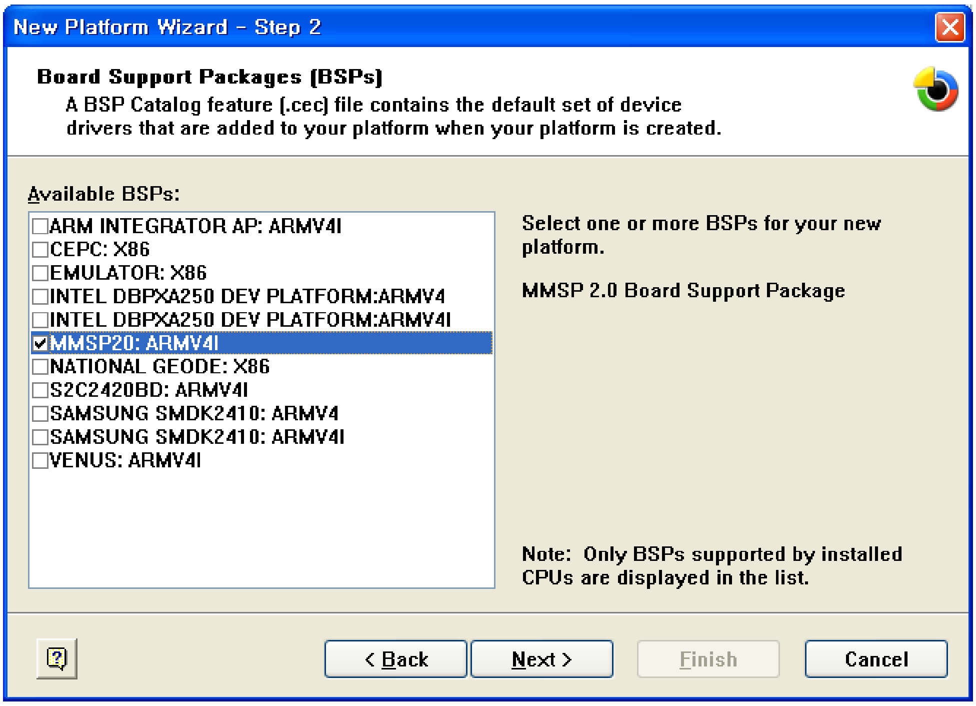
4. Register imported .cec file by pressing OK.
5. MP2520F BSP is added in catalog list.



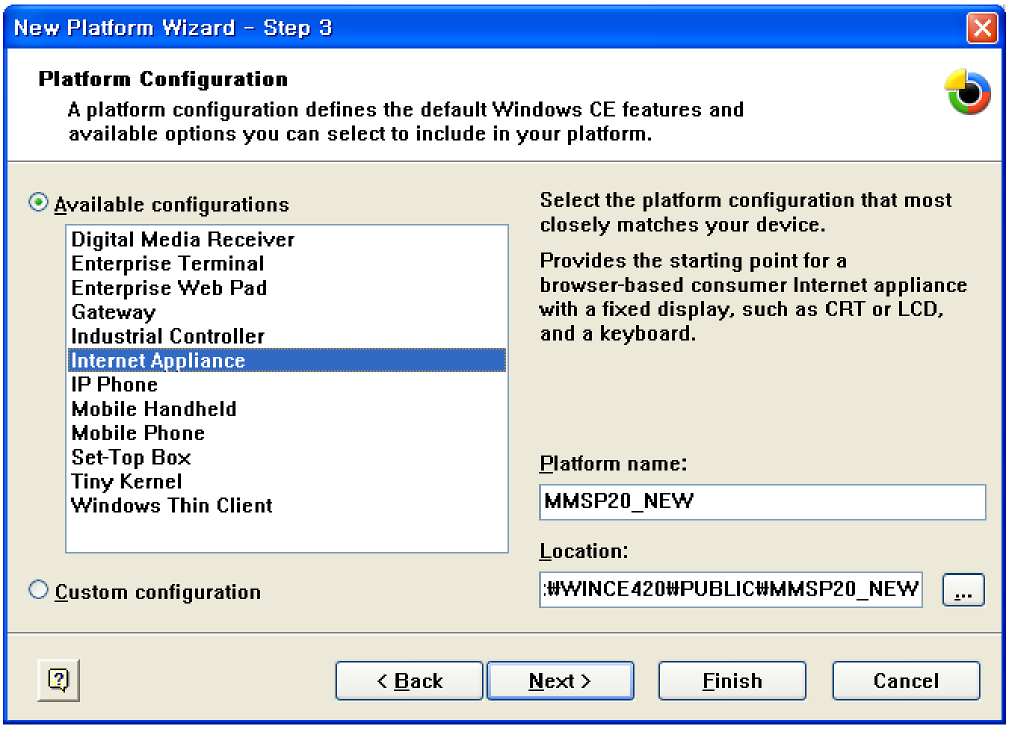
1.3 Creating New MMSP20 Platform

1. Go to **File** -> **New Platform** in Platform Builder menu.
2. Press **Next** and check MMSP20:ARMV4I in New Platform Wizard.

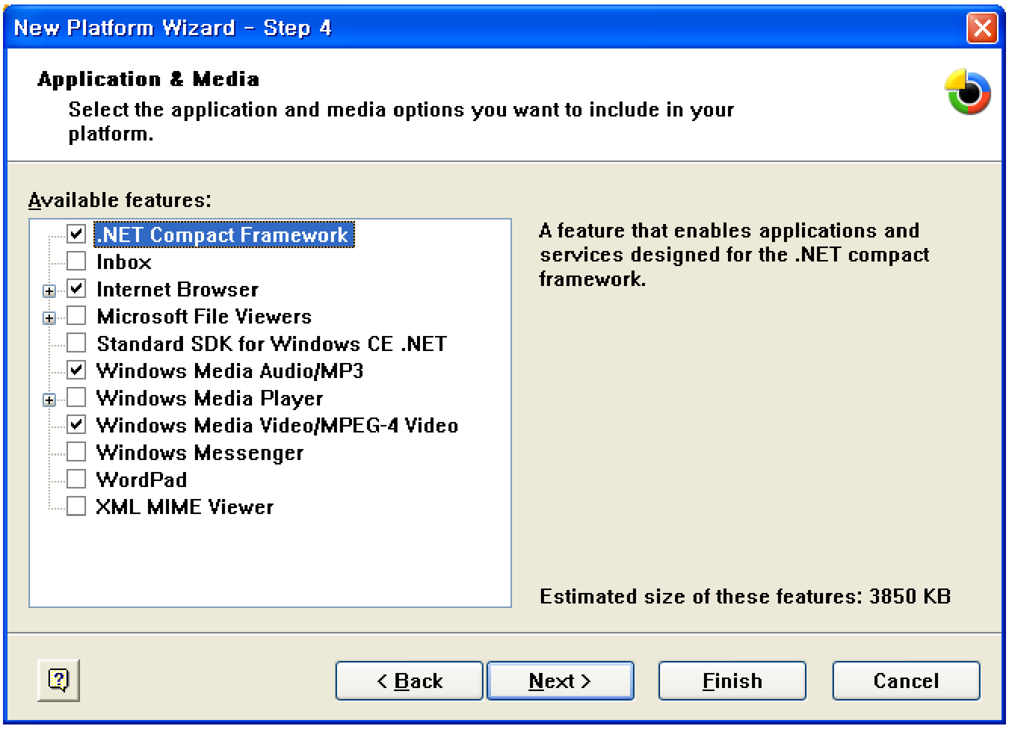




3. Select appropriate configuration and write name of the new platform.



4. Press **Next** > and check features needed. (Features shown in this window can be added or removed in Catalog window.)

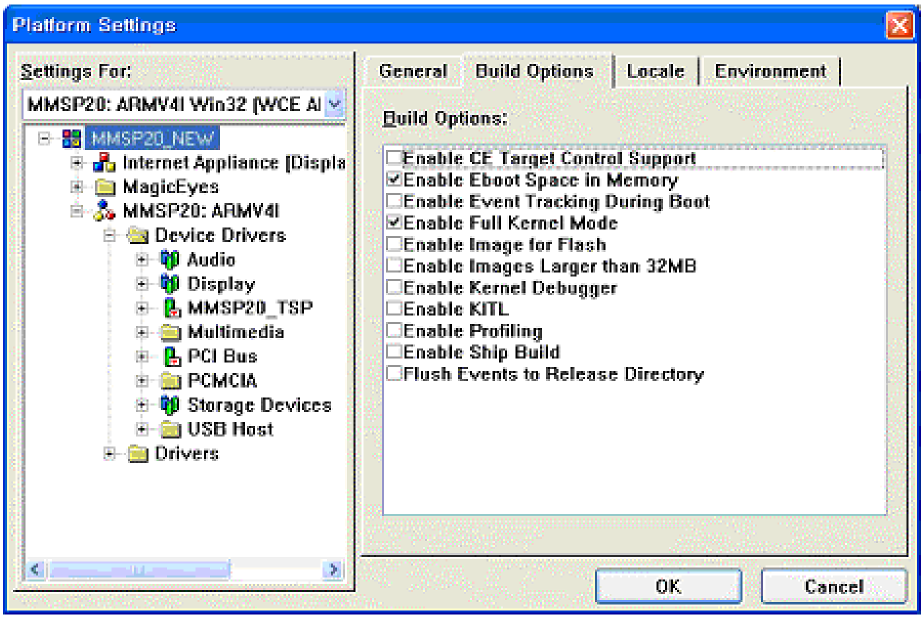


5. Complete New Platform creation by pressing **Finish** button.

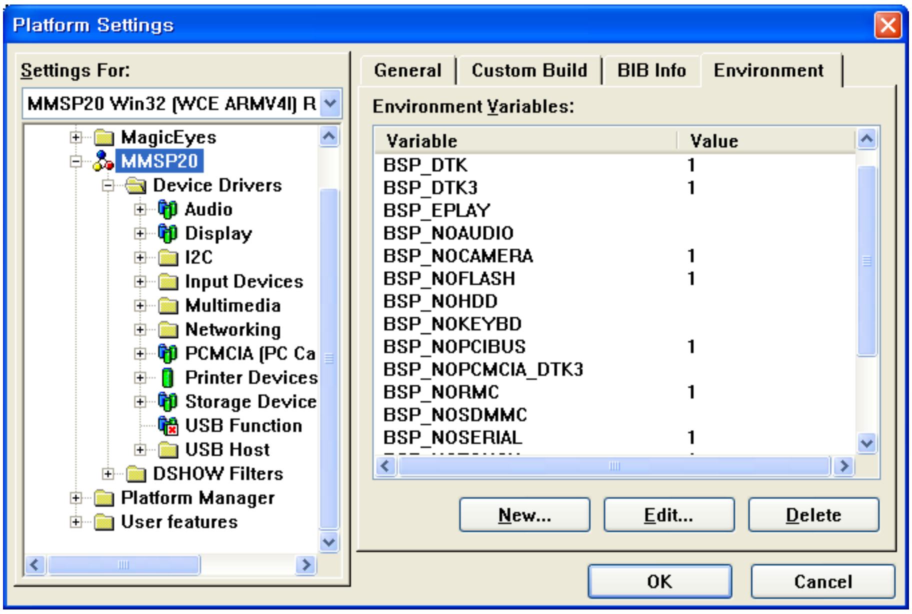
2. Platform Configuration

2.1 Configuration Settings for Development Board

1. Select **Platform** -> **Settings** in Platform Builder Menu.
2. Check **Enable Full Kernel Mode** in the **Build Options** list.



3. Select **Environment** tab and add or modify necessary variables.



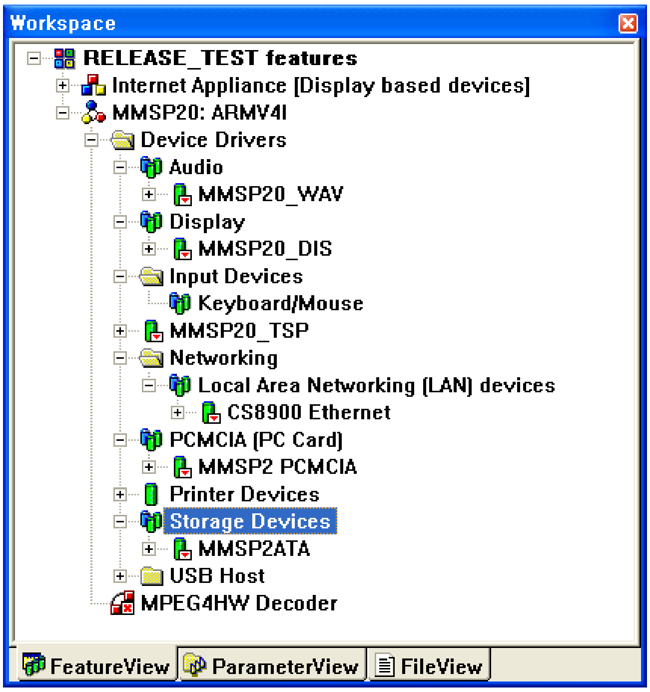
Variables that should be added are:

Variables H/W	BSP_DTK	BSP_DTK3	BSP_DTK4	BSP_PMP	BSP_EPLAY	BSP_SDRAM_128
Old DTK (before rev.3)	**1**					**1** (or null, if you want to use only 64MB)
Old DTK rev.3	**1**	**1**				**1** (or null, if you want to use only 64MB)
PMP2nd				**1**		
DTK rev.4.1.1			**1**			**1** (or null, if you want to use only 64MB)

4. Set DIP switch on the board appropriately. (See "Output Mode Selection" of the Display Driver Guide)

2.2 Adding Basic Components

Remove unnecessary drives shown in the MMSP20:ARMV4I → Device Drivers of the **FeatureView** window.



For example, remove Serial which is not supported by MP2520F BSP.

Add necessary components according to the driver guides in this document.

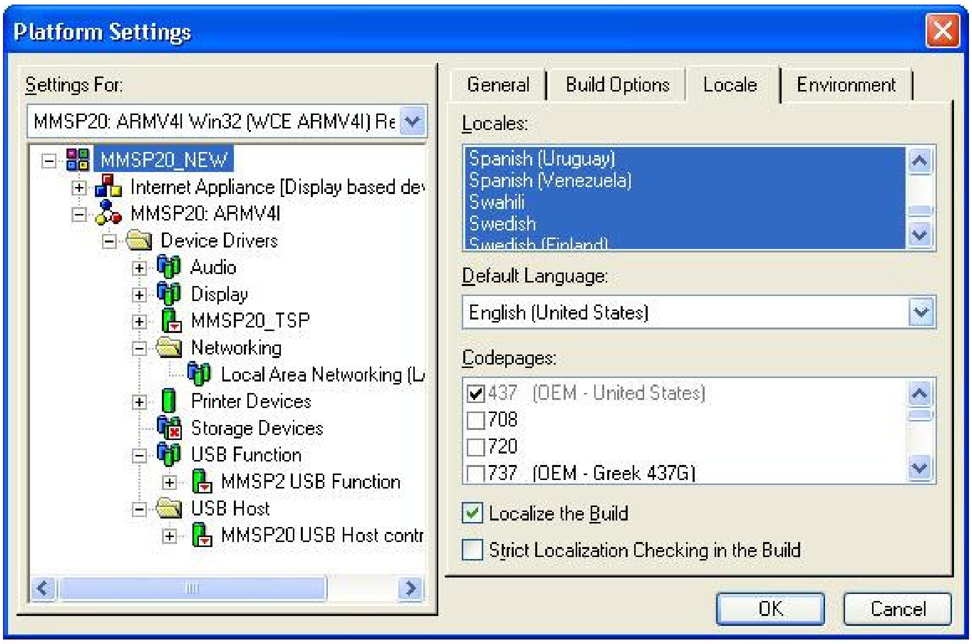
(**Errors can be occurred during build process when components required by each driver are not added and configured correctly. For example, Display Driver is built correctly when DirectDraw component is included.**)

For configuring the drivers and components, you must refer to each guide in chapter 3 through 11.

Default language of the Windows CE OS follows the language of desktop Windows that you have installed Platform Builder. If you installed Platform Builder on desktop Windows with Asian language (Korean, Japanese, Chinese, etc.), you can reduce image of the Windows CE OS by changing default language to English. To change default language:

Select **Platform** -> **Settings** Menu.

Select **Locale** and Change default language to English.



This is an example of Platform Builder installed on Windows XP.

3. Display Driver Guide

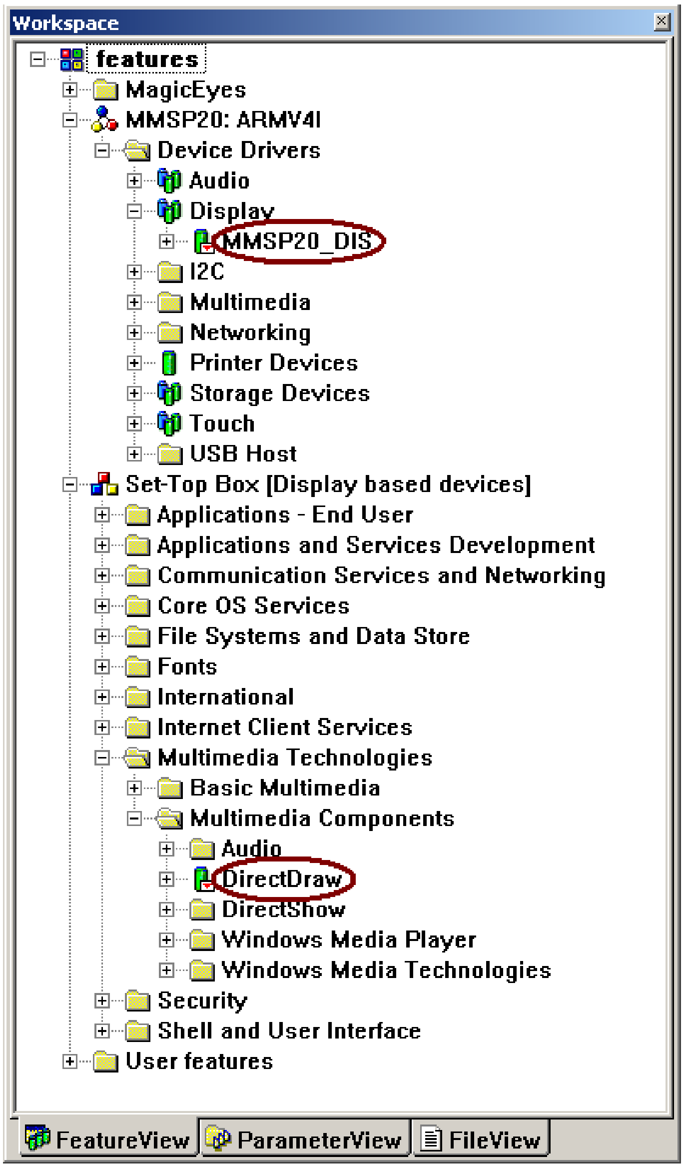
3.1 Adding Display Driver

3.1.1 Adding Display device driver

Add **MMSP20_DIS** component in **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** →**Display** of the catalog. (Click this item with right button of the mouse and select 'Add to Platform' menu.)

3.1.2 Adding DirectDraw component

Add **DirectDraw** component in **Core OS** → **Display Based Devices** → **Multimedia Technologies** → **Multimedia Components** of the catalog



3.2 Features

- Supports LCD & Video Encoder
- Supports 2 YUV H/W overlay surfaces.
- Supports Fine and coarse scale.
- Supports Bitblt H/W acceleration.
- Supports H/W Cursor.
- Supports OSD/SPU Layer
- Supports Color Controls

3.3 Limitations

- H/W Cursor is available up to 64x64.
- Two overlays cannot work in fine scale simultaneously.

3.4 Output Mode Selection

For DTK revision 4th or PMP 2nd

Connector	Resolution	SW2 (DIP Switch)				Remark
		4	3	2	1	
-	-	-	OFF	OFF	OFF	
D-SUB (Monitor)	640x480x60 Hz	-	OFF	OFF	ON	CX25874
D-SUB (Monitor)	800x600x60 Hz	-	OFF	ON	OFF	CX25874
Component(YPbPr)	720x480px60 Hz	-	OFF	ON	ON	CX25874
-	-	-	ON	OFF	OFF	
Composite/S-Video	720x480ix30 Hz	-	ON	OFF	ON	CX25874
LCD	800x600x60 Hz	-	ON	ON	OFF	
LCD	640x480x60 Hz	-	ON	ON	ON	

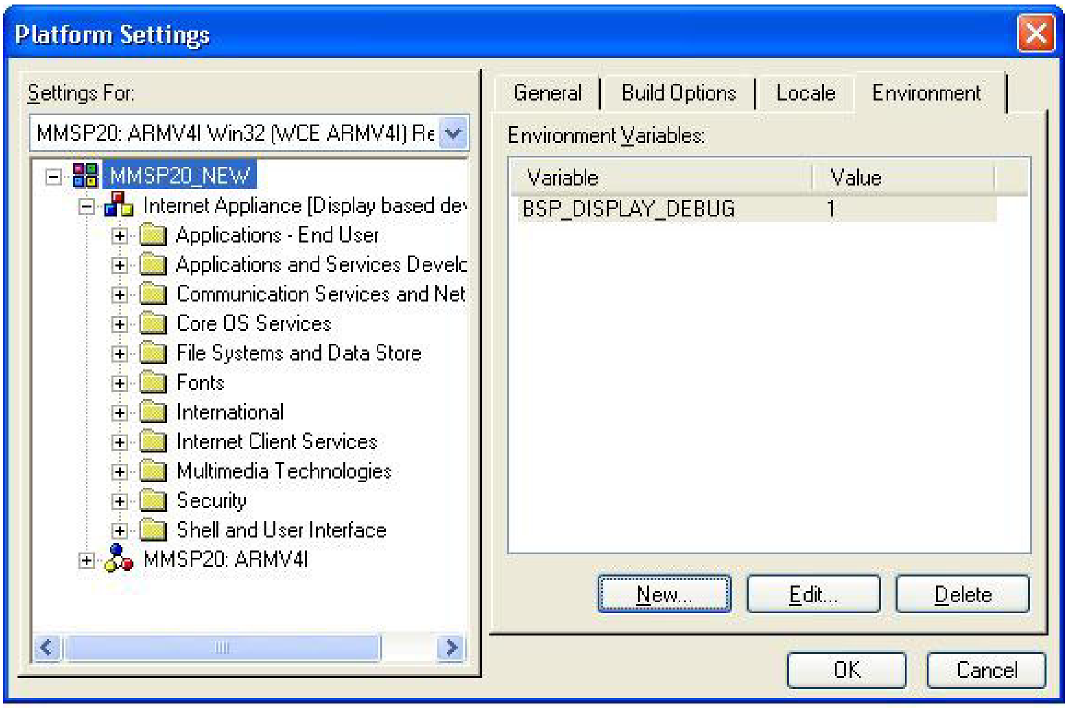
For DTK revision 3rd or earlier

Connector	Resolution	SW2 (DIP Switch)				Remark
		8	7	6	5 ~ 1	
-	-	OFF	OFF	OFF	-	
D-SUB (Monitor)	640x480x60 Hz	OFF	OFF	ON	-	CX25874
D-SUB (Monitor)	800x600x60 Hz	OFF	ON	OFF	-	CX25874
Component(YPbPr)	720x480px60 Hz	OFF	ON	ON	-	CX25874
-	-	ON	OFF	OFF	-	
Composite/S-Video	720x480ix30 Hz	ON	OFF	ON	-	CX25874
LCD	800x600x60 Hz	ON	ON	OFF	-	
LCD	640x480x60 Hz	ON	ON	ON	-	

※ A video option board is required to output D-SUB/Component/Composite

3.5 Debug Output

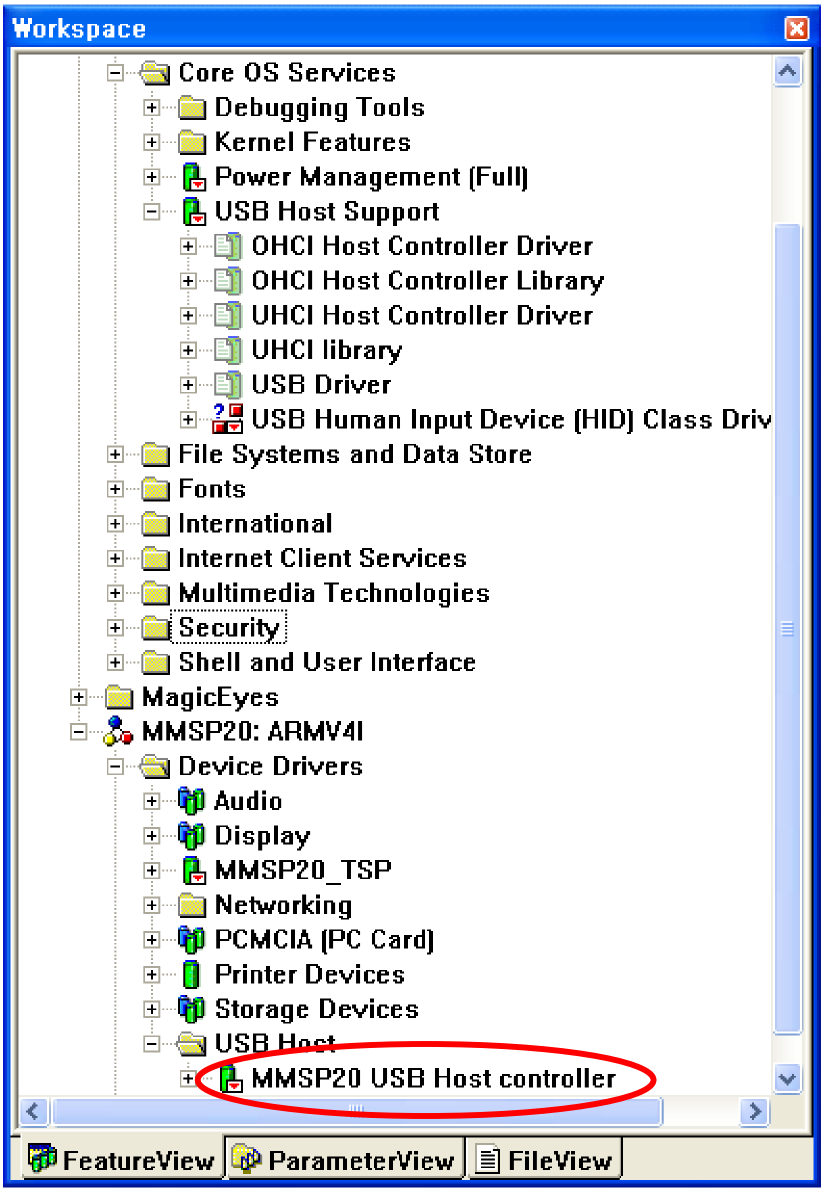
If you want to see display driver's debugging message in RETAIL mode, set **BSP_DISPLAY_DEBUG** environment value as 1



4. USB Host Driver Guide

4.1 Adding USB Host Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Host** → **USB Host Controllers** → **MMSP20 USB Host Controller** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



4.2 Using USB Mouse

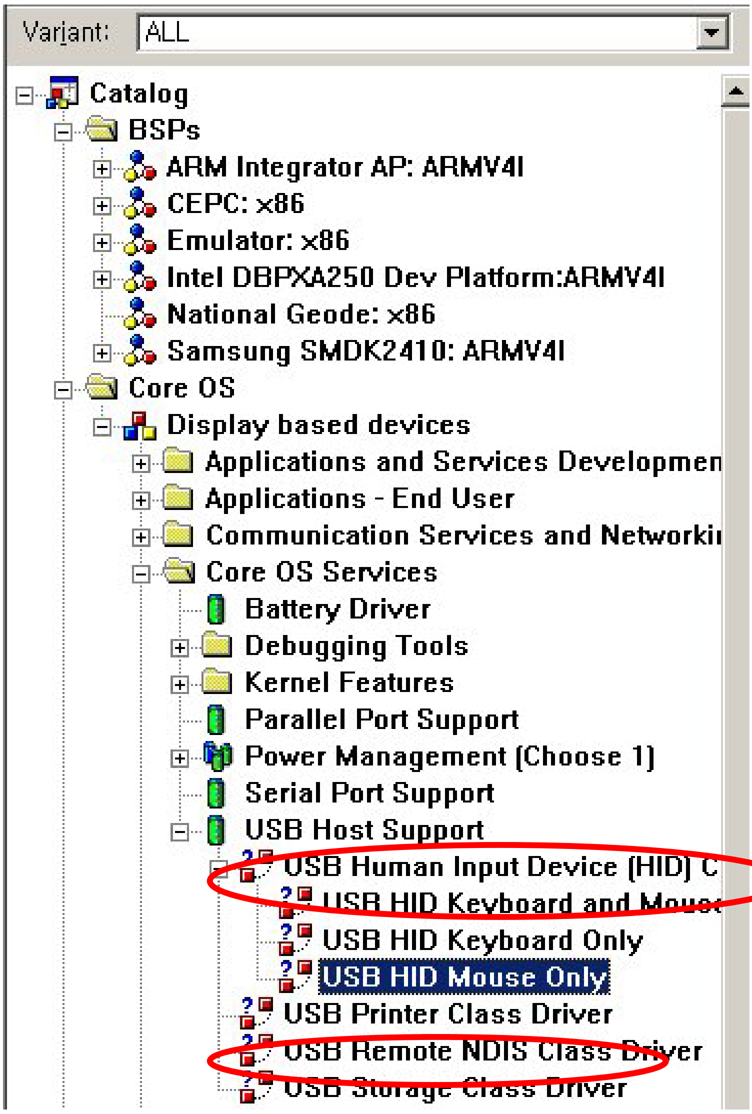
To use USB mouse, add **Core OS** → **Display based devices** → **Core OS services** → **USB Host Support** → **USB Human Input Device (HID) Class Driver** → **USB HID Keyboard and Mouse** component. (See picture below)

4.3 Using USB Keyboard

To use USB keyboard, add **USB HID Keyboard and Mouse** component and **Device Drivers -> Input Devices -> Keyboard/Mouse -> NOP(Stub) Keyboard/Mouse English** component.

4.4 Using USB Storage Device

To use USB storage devices, **add Core OS** → **Display based devices** → **Core OS services** → **USB Host Support** → **USB Storage Class Driver** component. (See picture below)


Variant: ALL
Catalog
BSPs
ARM Integrator AP: ARMV4I
CEPC: x86
Emulator: x86
Intel DBPXA250 Dev Platform:ARMV4I
National Geode: x86
Samsung SMDK2410: ARMV4I
Core OS
Display based devices
Applications and Services Developmen
Applications - End User
Communication Services and Networki
Core OS Services
Battery Driver
Debugging Tools
Kernel Features
Parallel Port Support
Power Management (Choose 1)
Serial Port Support
USB Host Support
USB Human Input Device (HID) C
USB HID Keyboard Only
USB HID Mouse Only
USB Printer Class Driver
USB Remote NDIS Class Driver
USB Storage Class Driver

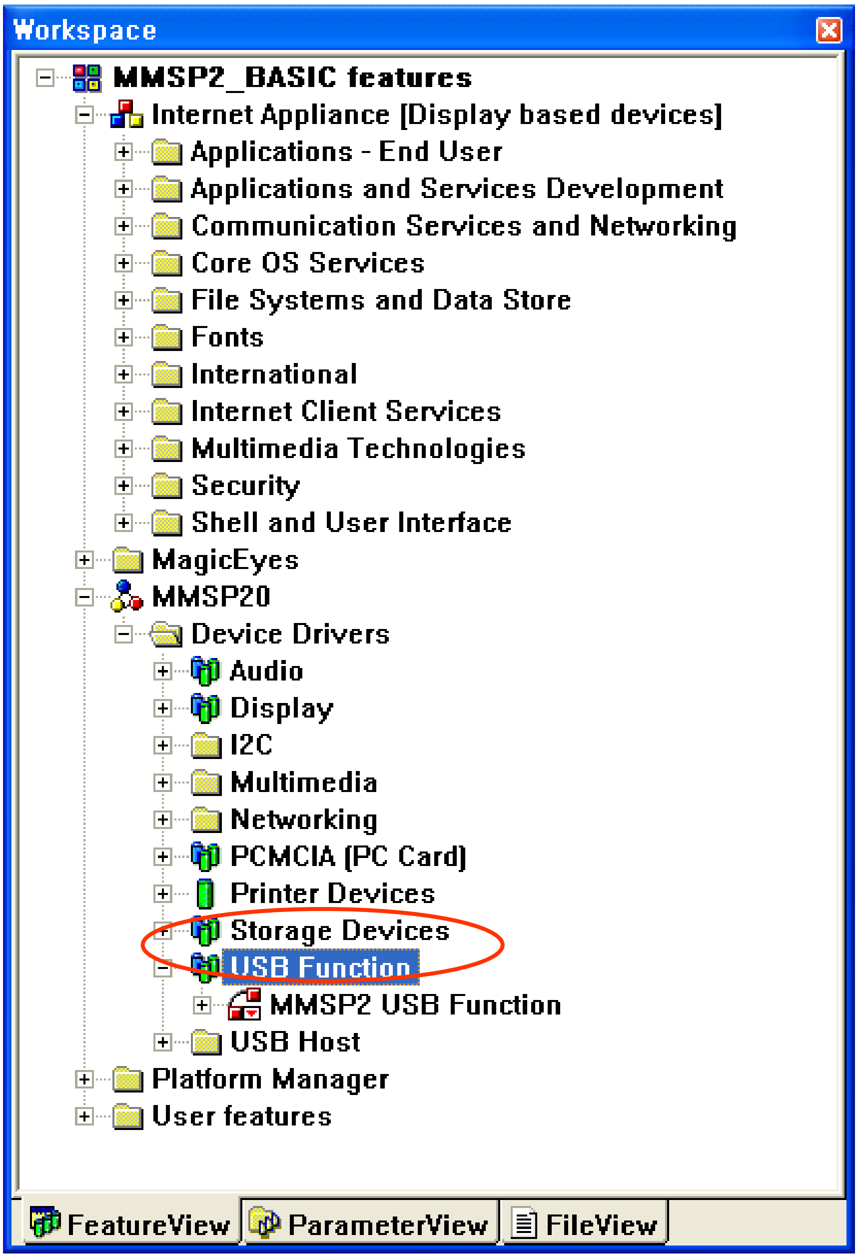
4.5 Using Virtual Keyboard

To use Virtual Keyboard instead of USB keyboard, **add Core OS → Shell and User Interface → User Interface → Software Input Panel → Software-based Input Panel [SIP] [Choose 1 or more] → SIP for Large/Small Screens** component.

5. USB 1.1 Function Driver Guide (on Chip)

5.1 Adding USB 1.1 Function Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Function** → **MMSP20 USB Function** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



5.2 Installing Device Driver on PC

Device driver for MMSP2 USB 1.1 is located at MMSP20\DRIVERS\USB\FUNCTION as wceusbsh.sys.

5.3 Using ActiveSync with USB Connection

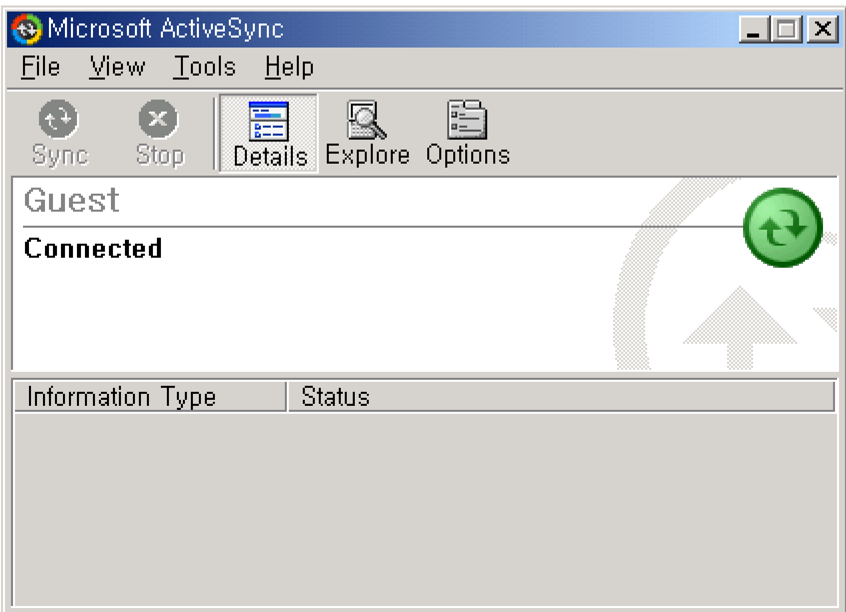
To test ActiveSync on DTK board, make sure things below are done.

- USB connection cable is prepared. (One side is A type, and the other side B type)
- ActiveSync application is installed on host PC.
- When you set configuration of new project with **New Platform Wizard**, Check **ActiveSync** feature.
- Build image with above configuration and other settings shown on this BSP guide.

After Windows CE image is built. Follow steps shown below to establish ActiveSync connection with host PC.

1. Load Windows CE image on DTK board. (Do not connect USB cable yet)
2. Open **Control Panel** of the Windows CE.
3. Run **'Network and Dial-up Connections'** of Control Panel.
4. Select **'Make New Connection.'**
5. Select **'SC2410 USB Cable'** and press **'Finish'** button.
6. Run **'PC Connection'** of Control Panel.
7. Press **'Change...'** button.
8. Select 'My Connection.' (or other name you given when you created new connection)
9. Press OK button to exit.
10. Connect USB cable.
11. Try to remove and re-connect USB cable when ActiveSync connection is not established correctly.

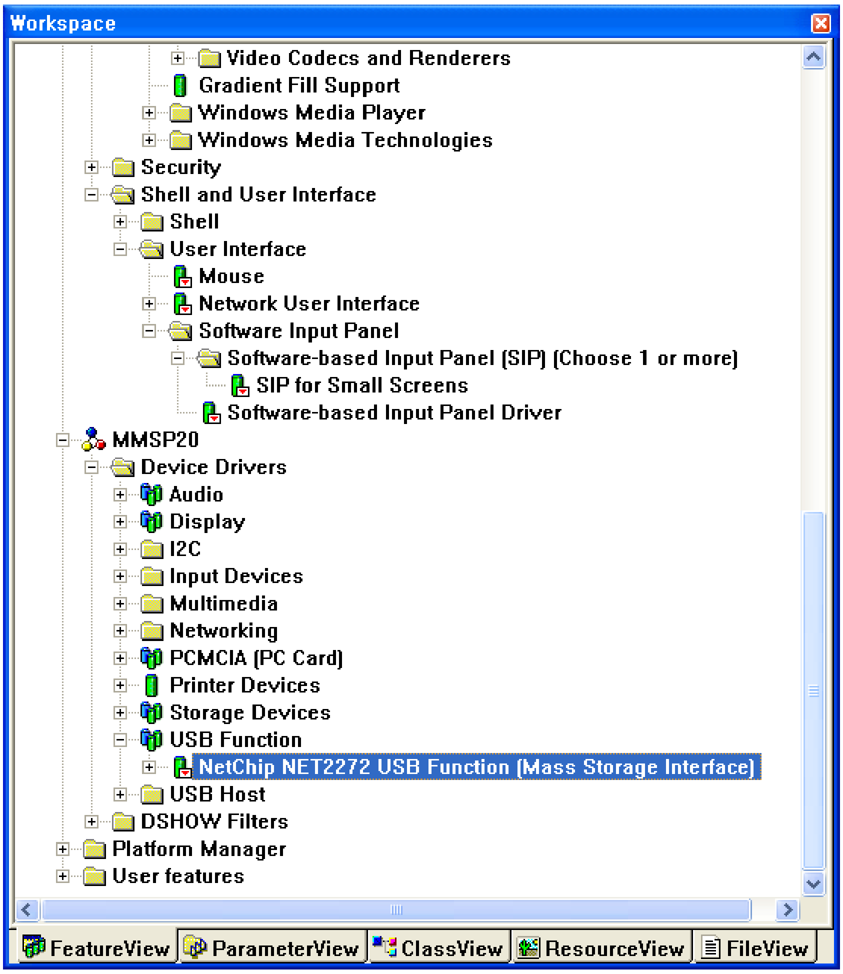
When some settings are completed, ActiveSync window will be shown like this:



6. USB 2.0 Mass Storage Driver Guide

6.1 Adding USB 2.0 Mass Storage Driver Component

Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **USB Function** → **NetChip NET2272 USB Function (Mass Storage Interface)** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



6.2 Connecting with Host

When USB 2.0 mass storage driver is initialized correctly, connect USB cable with host system. (e.g. desktop PC)

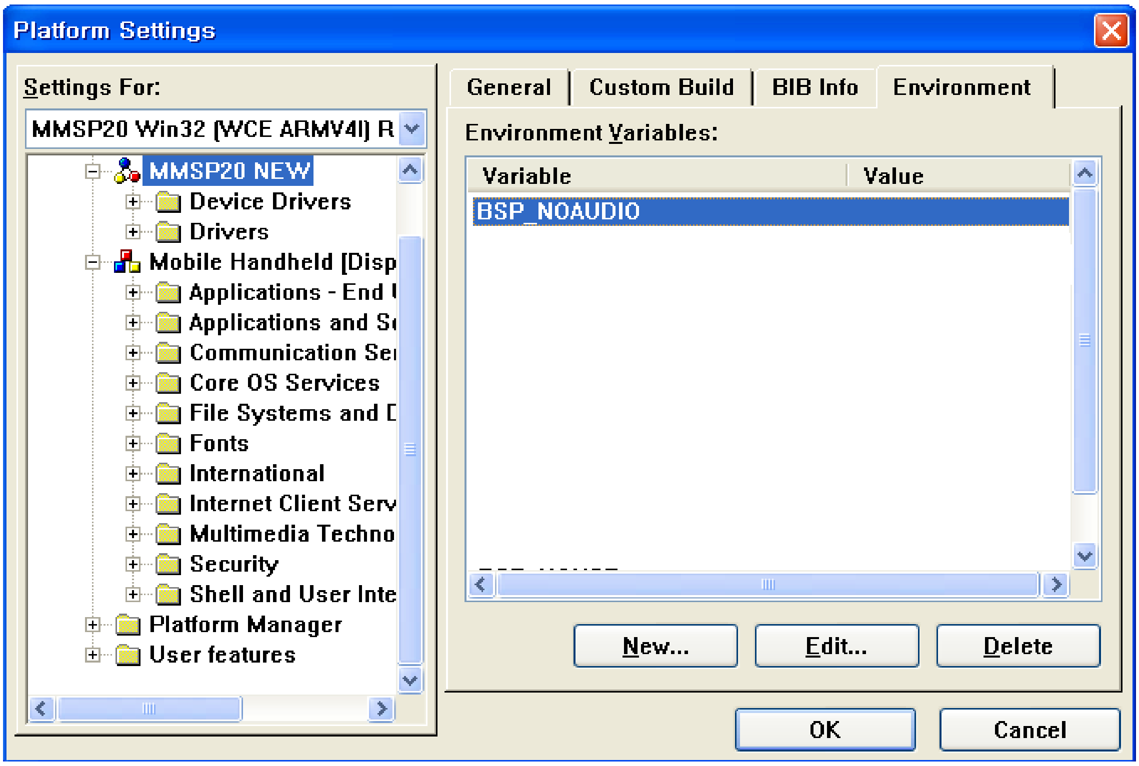
- On the DTK 4^{th} board, use left one of the two USB device connectors.

After connection is established, MMSP2 based system is mounted as a removable disk on the host side.

7. AC97 Driver Guide

7.1 Environment Variable Setting

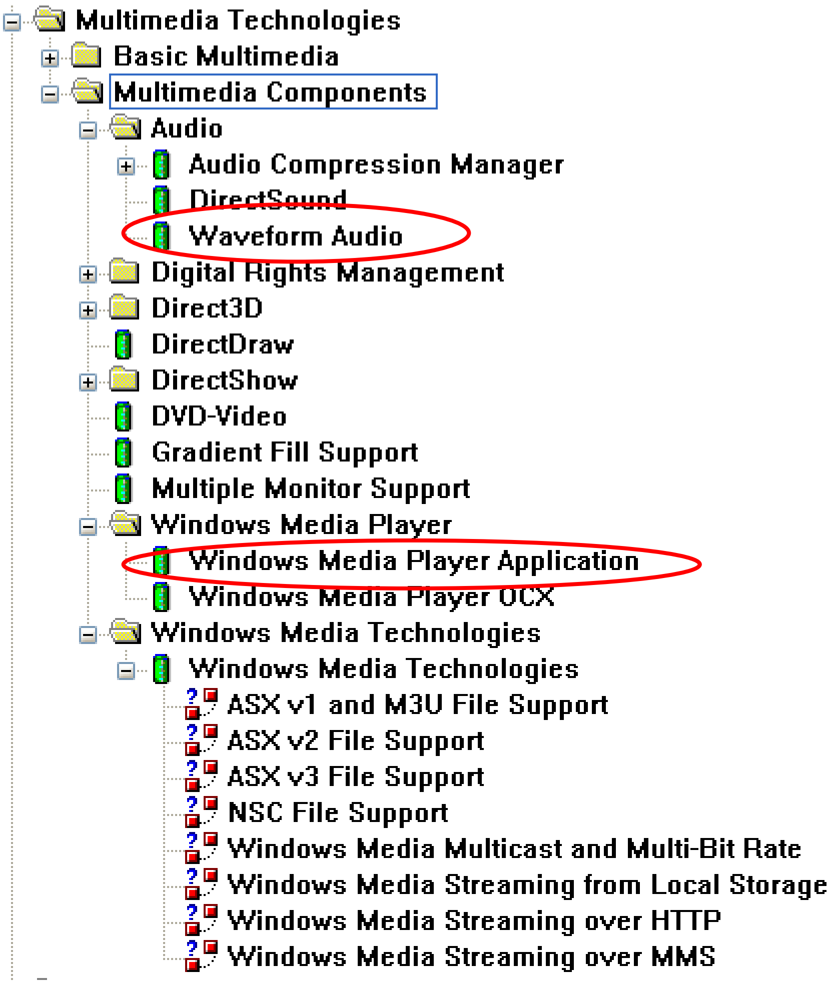
Press Setting in menu Platform, Set BSP_NOAUDIO environment value to NULL (blank) if it exists. If this value is 1, audio device driver cannot be used.



Note : If you cannot see BSP-NOAUDIO in Environment Variables in Environment, you have input "BSP_NOAUDIO" in environment after press "New…" and Press "OK"

7.2 Adding Required Components

- Add **Third Party** → **BSPs** → **MMSP20:ARMV4I** → **Device Drivers** → **Audio** → **MMSP20_WAV** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)
- Add **Core OS -> Display Based Devices -> Multimedia Technologies -> Multimedia Components -> Audio -> Waveform Audio** component for basic waveform audio play. (Click this item with right button of the mouse and select 'Add to Platform' menu.)
- Add **Core OS -> Display Based Devices -> Multimedia Technologies -> Multimedia Components -> Windows Media Player -> Windows Media Player Application** as an audio player application (**Windows Media Player OCX** also)



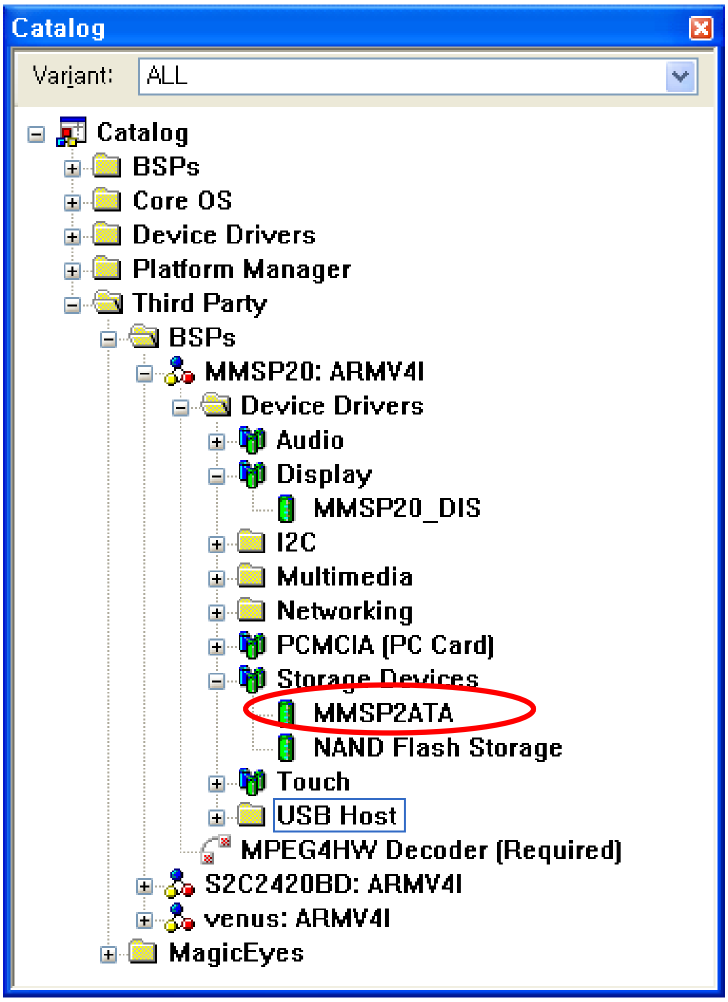
7.3 Adding Optional Components

To play other format of audio files, appropriate components should be added.

8. ATA Driver Guide

8.1 Adding ATA Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **MMSP2ATA** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)

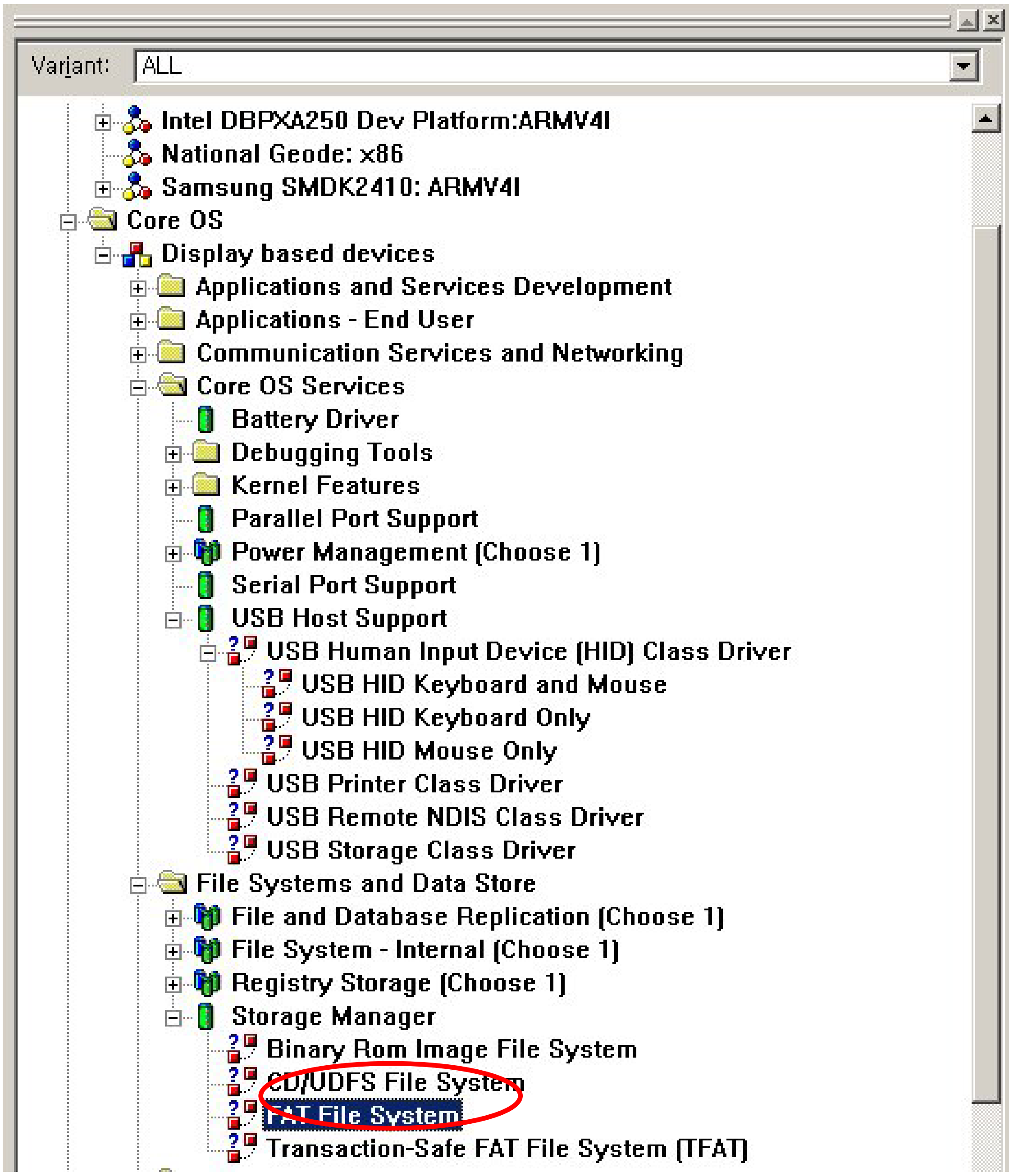


8.2 Features

- Supports PIO input/output of the ATA Hard Disk.

8.3 Adding Required Components

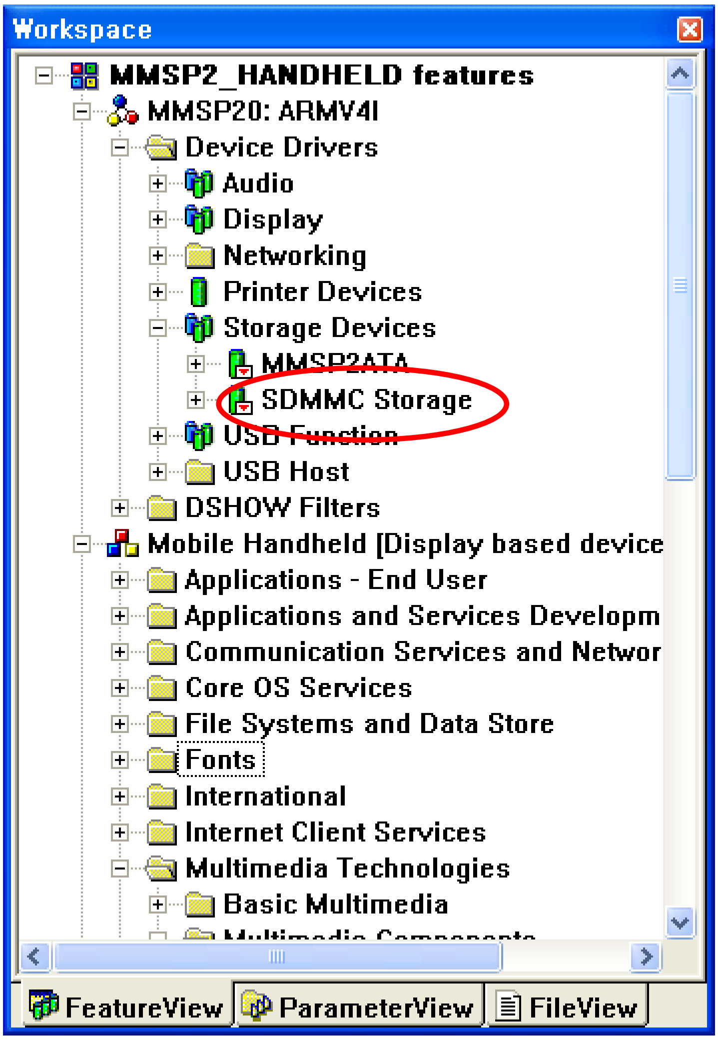
- Add **Core OS** → **Display based devices** → **File System and Data Store** → **Storage Manager** → **FAT File System** Component so that disk can be formatted as FAT file system. (See picture)


Variant: ALL
Intel DBPXA250 Dev Platform:ARMV4I
National Geode: x86
Samsung SMDK2410: ARMV4I
Core OS
Display based devices
Applications and Services Development
Applications - End User
Communication Services and Networking
Core OS Services
Battery Driver
Debugging Tools
Kernel Features
Parallel Port Support
Power Management (Choose 1)
Serial Port Support
USB Host Support
USB Human Input Device (HID) Class Driver
USB HID Keyboard and Mouse
USB HID Keyboard Only
USB HID Mouse Only
USB Printer Class Driver
USB Remote NDIS Class Driver
USB Storage Class Driver
File Systems and Data Store
File and Database Replication (Choose 1)
File System - Internal (Choose 1)
Registry Storage (Choose 1)
Storage Manager
Binary Rom Image File System
CD/UDFS File System
FAT File System
Transaction-Safe FAT File System (TFAT)

9. SD/MMC Storage Driver Guide

9.1 Adding SD/MMC Storage Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **SDMMC Storage** component. (Click this item with right button of the mouse and select 'Add to Platform' menu.)



9.2 Features

- Supports Secure Digital (SD) Storage Card.

9.3 Limitations

- Multimedia Card (MMC) is not supported.

9.4 Adding Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **ATA Drive** section)

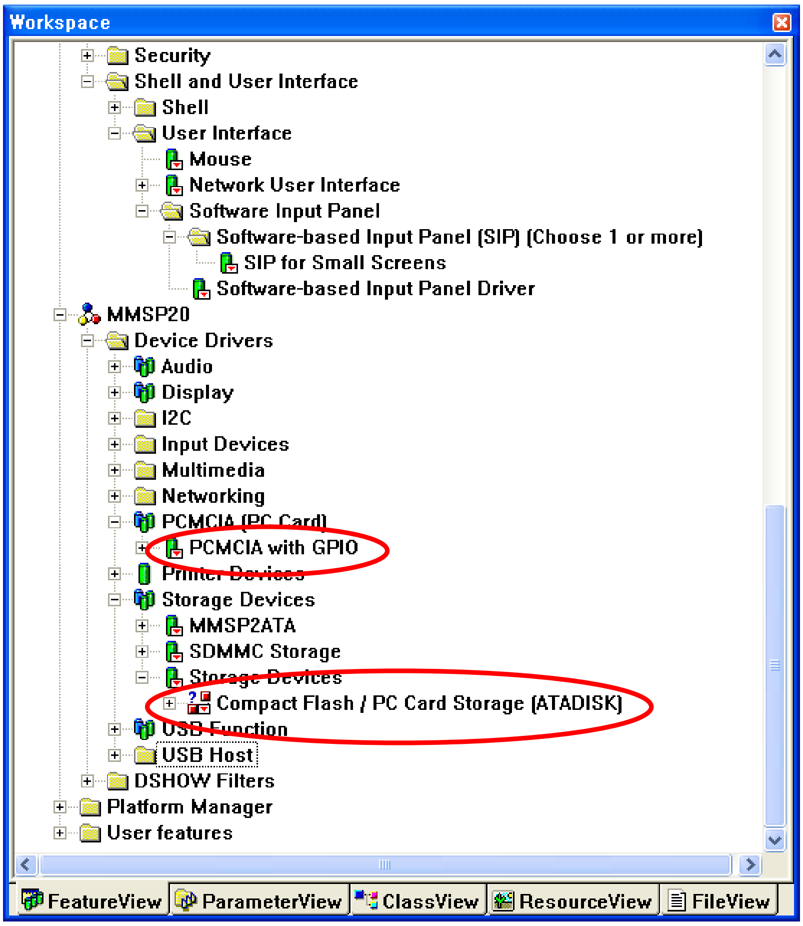
10. CompactFlash Storage Driver Guide

10.1 Adding CompactFlash Storage Device Driver

CompactFlash storage driver runs on the CompactFlash/PCMCIA bus driver layer. So you must include both CompcatFlash storage and CompactFlash/PCMCIA bus driver.

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **PCMCIA (PC Card)** → **PCMCIA_GPIO** component. (This is the CompactFlash / PCMCIA bus driver.)

Add **Device Drivers** → **Storage Devices** → **Storage Devices** → **CompactFlash / PC Card Storage (ATADISK)** component. (This is the CompactFlash storage driver.)



10.2 Supported Platforms

CompactFlash/PCMCIA storage is can be used on DTK 4th board only.

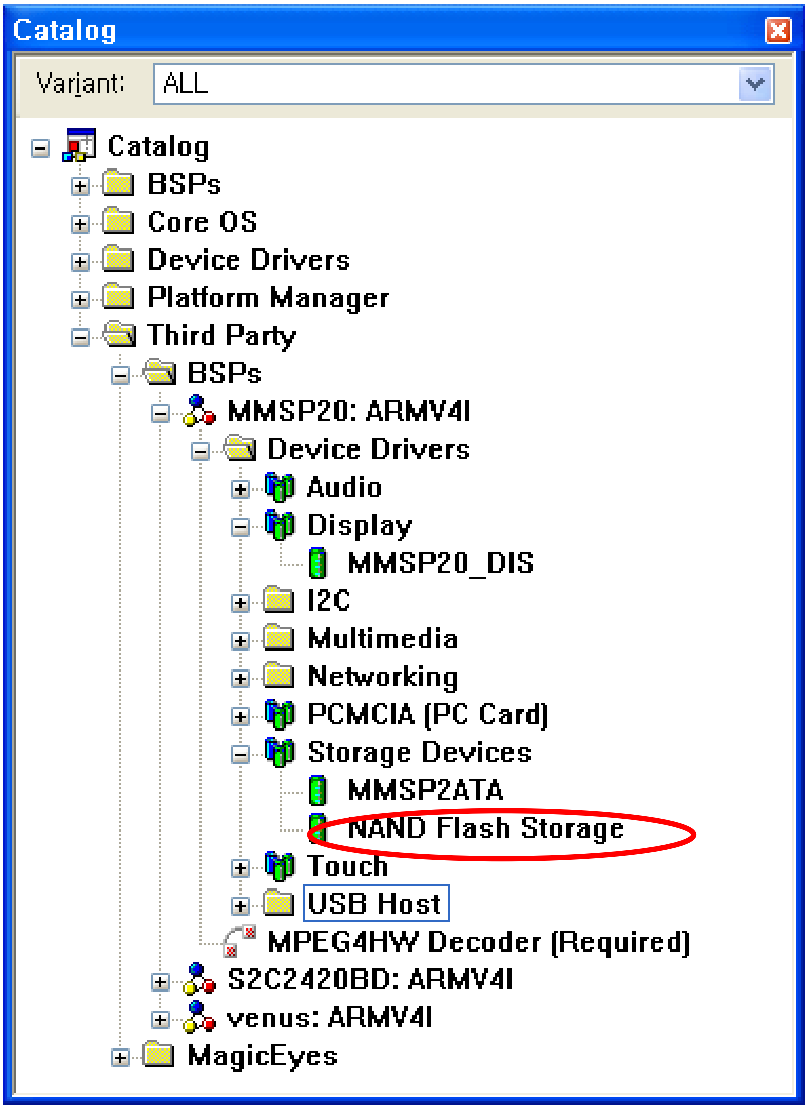
10.3 Adding Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **ATA Drive** section)

11. NAND Flash Storage Driver Guide

11.1 Adding NAND Flash Storage Device Driver

Add **Third Party** → **BSPs** → **MMSP20: ARMV4I** → **Device Drivers** → **Storage Devices** → **NAND Flash Storage** component.



11.2 Required Components

- Add FAT File System Component so that disk can be formatted as FAT file system. (See 'Required Components' of the **Chapter 7. ATA Drive** section)

12. Camera Driver Guide

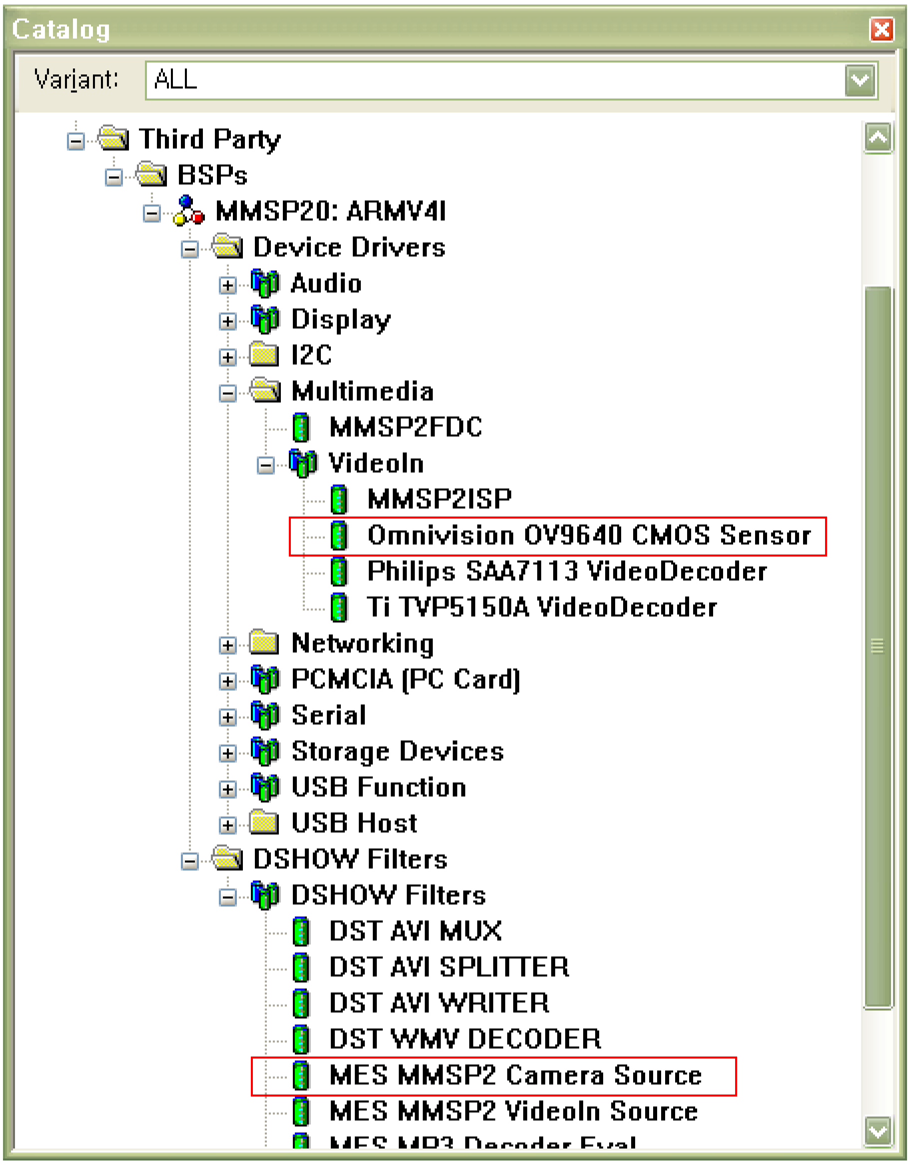
12.1 Adding Camera Driver

A. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn**

B. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **MMSP2ISP**

C. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **Omnivision OV9640 CMOS Sensor**

※ For DTK revision 3rd or earlier, a camera option board is required. For PMP2, DTK4 and DTK4.1.1 board, a camera module is required.



12.2 Adding Related Features

A. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MMSP2 Camera Source** component.

B. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MPEG4 Video Encoder** component.

C. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI MUX** component.

D. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI WRITER** component.

Add DirectShow related features in Core OS -> Multimedia Technologies. See **MPEG4HW Encoder Filter Guide** section for more information.

12.3 Running Camera Test Application

When Camera component is included, a camera test application is also included in OS image. You can run this application on Windows CE Explorer. Open Explorer and change folder to \Windows, then run '**Camera**'.

13. Video Decoder Driver Guide

13.1 Adding Video Decoder Driver

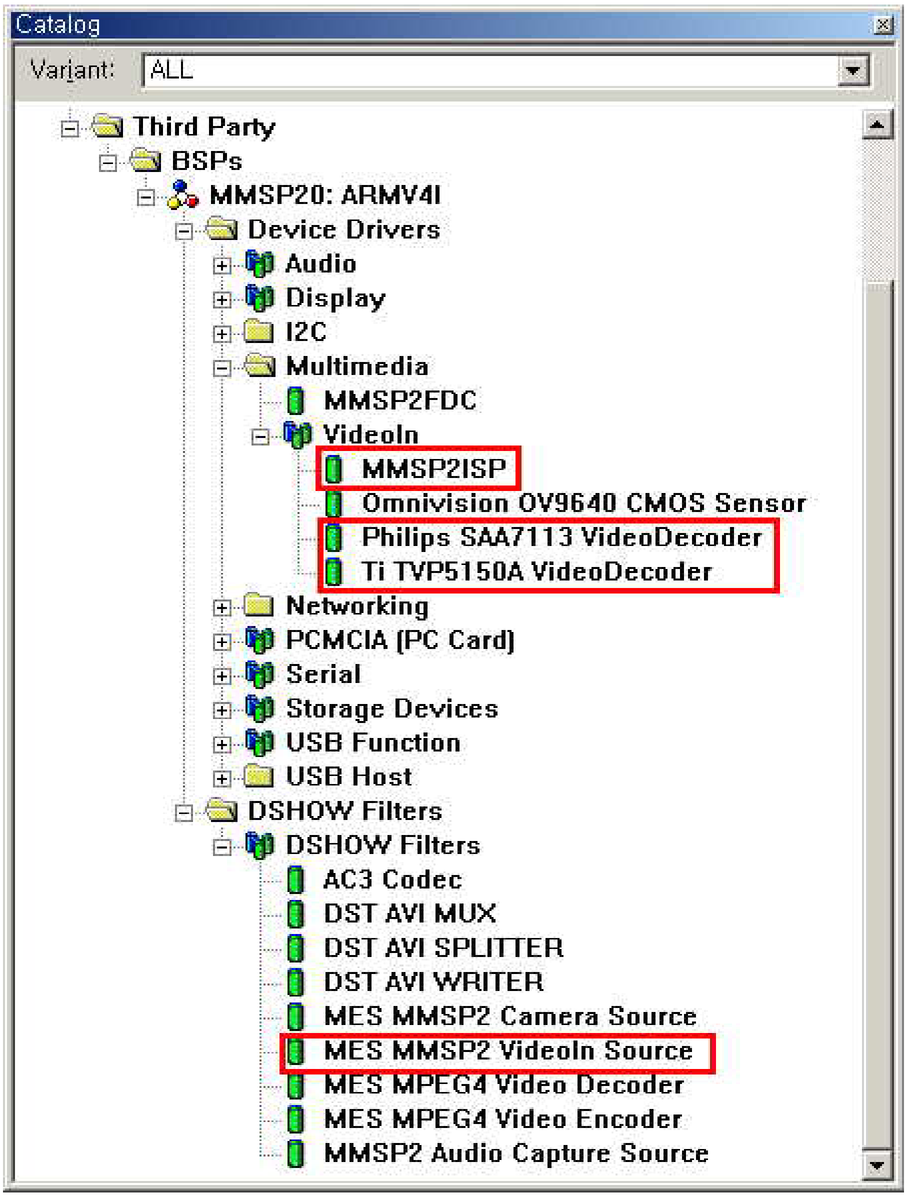
A. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn**

B. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **MMSP2ISP**

C. Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Multimedia** → **VidoeIn** → **Philips SAA7113 VideoDecoder or Ti TVP5150A VideoDecoder.**

※ For DTK revision 4th or PMP 2nd, select Philips SAA7113 VideoDecoder.

※ For DTK revision 3rd or earlier, a video decoder option board is required.



13.2 Adding Related Features

A. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MMSP2 VideoIn Source** component.

B. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **MES MPEG4 Video Encoder** component.

C. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI MUX** component.

D. Add **Third Party** → **BSPs** → **MMSP20** → **DSHOW Filters** → **DSHOW Filters** → **DST AVI WRITER** component.

Add DirectShow related features in Core OS -> Multimedia Technologies. See **MPEG4HW Encoder Filter Guide** section for more information.

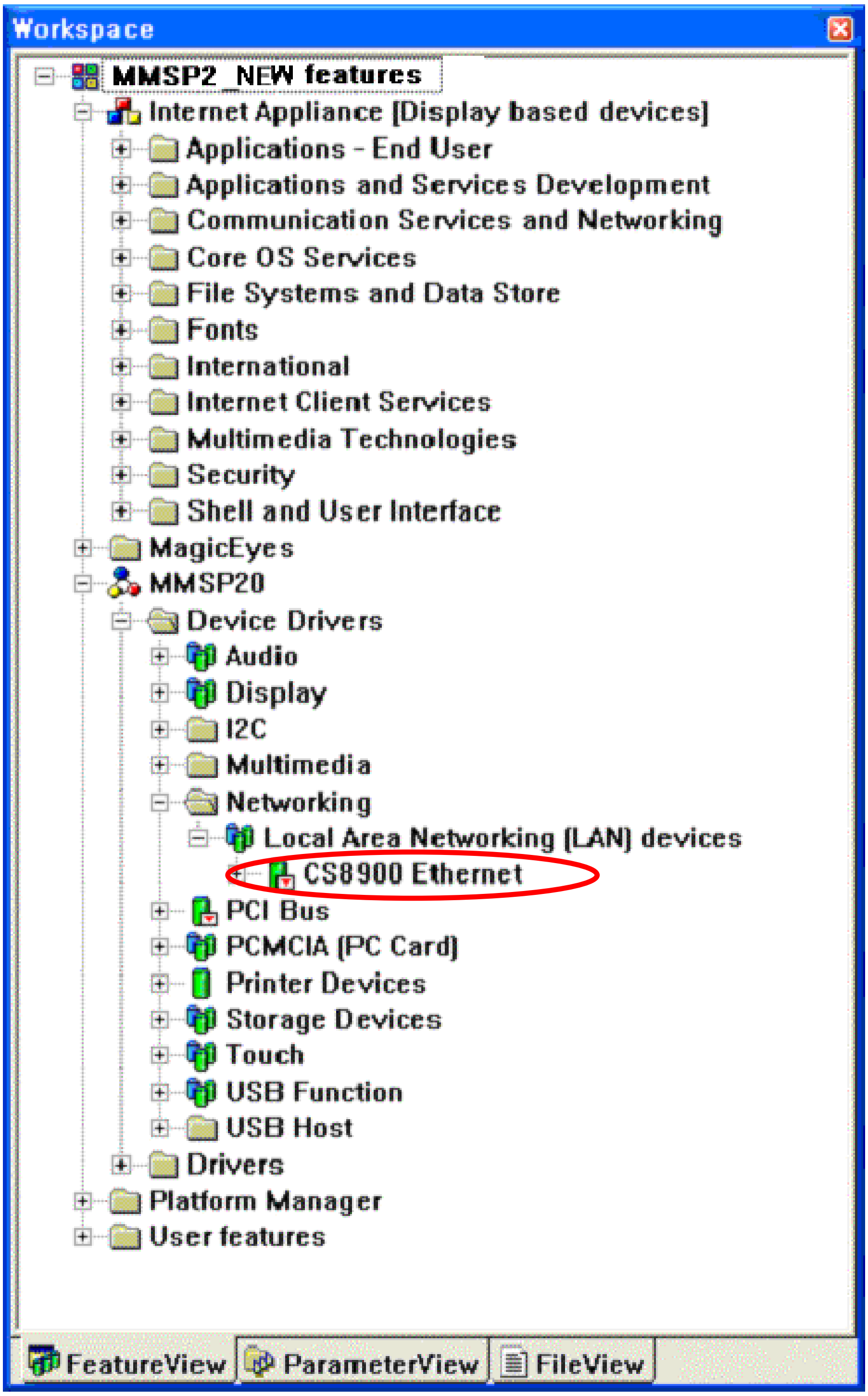
13.3 Running VideoIn Test Application

When Video decoder component is included, a VideoIn test application is also included in OS image. You can run this application on Windows CE Explorer. Open Explorer and change folder to \Windows, then run '**VideoIn**'.

14. Local Area Networking Controller Driver (CS8900) Guide

14.1 Adding CS8900 Driver

Add **Third Party** → **BSPs** → **MMSP20** → **Device Drivers** → **Networking** → **Local Area Networking** → **CS8900 Ethernet** component.



Open **Workspace – ParameterView – MMSP20_NEW parameters - C:\WINCE420 - MMSP20 - Project Specific Files** - **project.reg** in the ParameterView and add registry contents shown below.

```
IF BSP_NOETHER !
IF BSP_NIC_CS8900
; @CESYSGEN IF BSP_NIC_CS8900
#include "$(_WINCEROOT)\PLATFORM\MMSP20\DRIVERS\NETCARD\CS8900\ENDS4ISA.reg"
; @CESYSGEN ENDIF BSP_NIC_CS8900
ENDIF BSP_NIC_CS8900
ENDIF BSP_NOETHER !
```

14.2 Available Features with Ethernet Driver

- Connect to Web by using Internet Explorer.
- Files in DTK board can be accessed when FTP Server component is included.

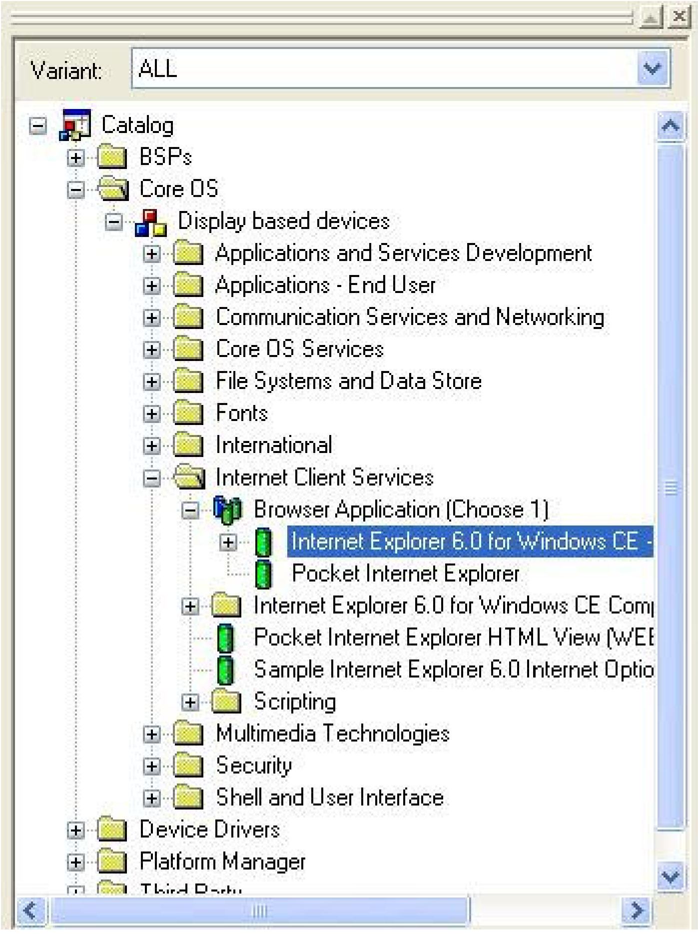
14.3 Setting IP Address

To set IP address, open **ends4isa.reg** in **WINCE420\PLATFORM\MMSP20\DRIVERS\NETCARD\CS8900** folder and modify value marked as bold text.

```
[HKEY_LOCAL_MACHINE\Comm\ENDS4ISA1\Parms\TcpIp]
    ; This should be MULTI_SZ
    "DefaultGateway"="220.78.49.252"
    ; This should be SZ... If null it means use LAN, else WAN and Interface.
    "LLInterface"=""
    ; Use zero for broadcast address? (or 255.255.255.255)
    "UseZeroBroadcast"=dword:0
    ; Thus should be MULTI_SZ, the IP address list
    "IpAddress"="220.78.49.136"
    ; This should be MULTI_SZ, the subnet masks for the above IP addresses
    "Subnetmask"="255.255.255.128"
    "DNS"="168.126.63.1"
    "EnableDHCP"=dword:0
```

14.4 Networking Related Components

- Select one of IE 6.0 or Pocket IE in **Internet Client Services** → **Browser Application** to use Internet Explorer.



- Add **FTP Server** component in **Communication Services and Networking** → **Servers** of the catalog. Add registry entry shown below to the project.reg to enable anonymous login to FTP server.

```
; @CESYSGEN IF SERVERS_MODULES_FTPD
; @CESYSGEN IF SERVERS_MODULES_SERVICES
[HKEY_LOCAL_MACHINE\Services\FTPD]
; @CESYSGEN ELSE
; @CESYSGEN ENDIF SERVERS_MODULES_SERVICES
    "Dll"="FTPD.Dll"
    "Order"=dword:9
    "Keep"=dword:1
    "Prefix"="FTP"
```

```
    "Index"=dword:0

[HKEY_LOCAL_MACHINE\COMM\FTPD]
     "IsEnabled"=dword:1
     "UseAuthentication"=dword:1
;    "UserList"="add;semicolon;separated;list;of;users;here"
     "AllowAnonymous"=dword:1
     "AllowAnonymousUpload"=dword:1
     "AllowAnonymousVroots"=dword:1
     "DefaultDir"="\\"
;To control logging
     "DebugOutputChannels"=dword:2
     "DebugOutputMask"=dword:17
     "BaseDir"="\\Windows"
     "LogSize"=dword:1000
; @CESYSGEN ENDIF SERVERS_MODULES_FTPD
```

15. Button Device Driver

Button device driver gets key input and let a application know what key is pressed. The source code is located at **MMSP20/DRIVERS/REMOTECTLR**. And we provide a simple application to test key input (**MMSP20/APPS/ ButtonTest**)

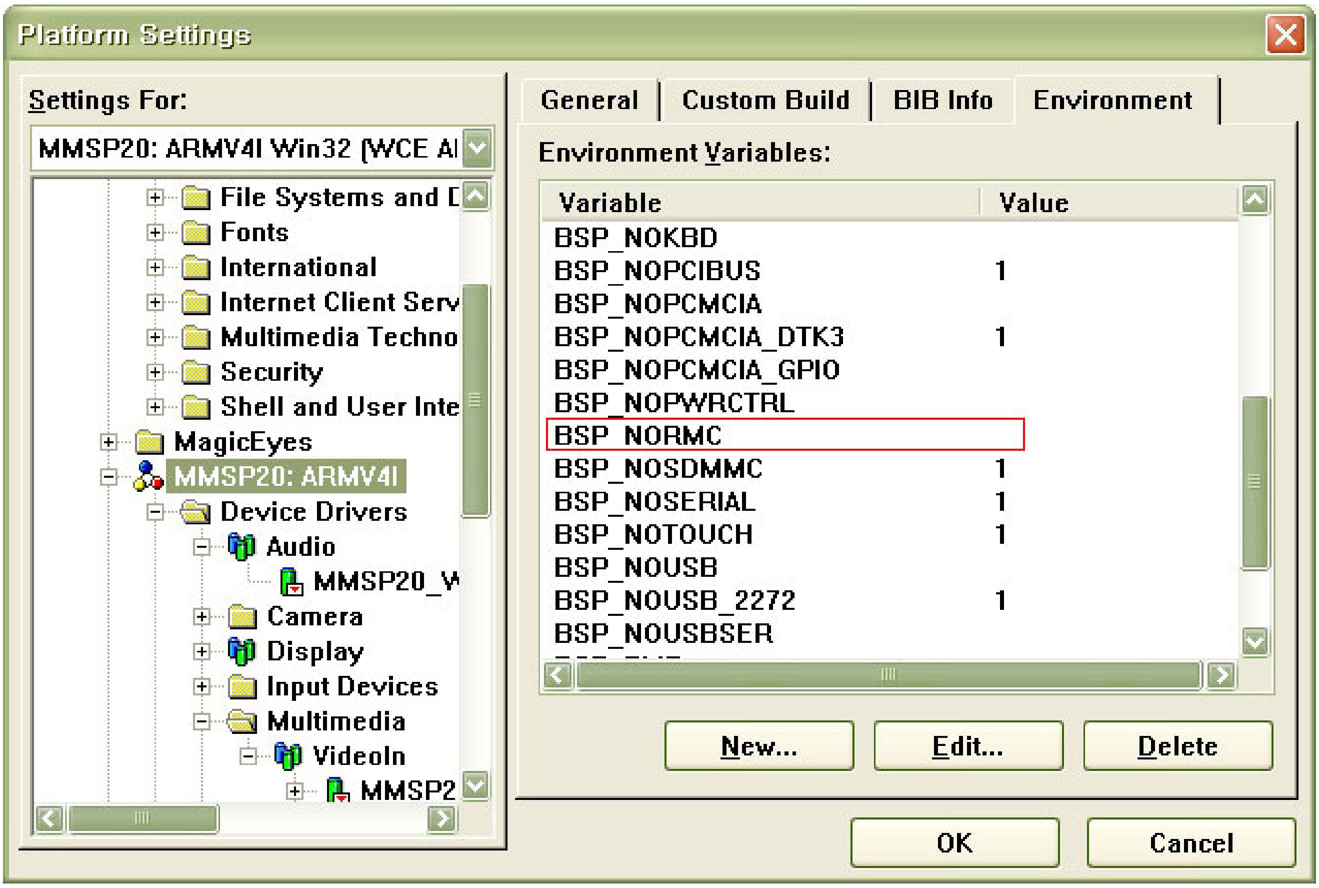
Button device driver supports **PMP2** and **DTK4** development kit that has 5 general purpose keys (SW3 ~ SW7).

Key Assignments	
SW3	UP (VK_UP)
SW4	DOWN (VK_DOWN)
SW5	MENU (VK_MENU_
SW6	EXIT (VK_ESCAPE)
SW7	ENTER (VK_RETURN)

When you press a key it generates a KEY event.

15.1 Adding Button Device Driver

To add button device driver, you have to set BSP_NORMC variable as null.



15.2 Known problem(s)

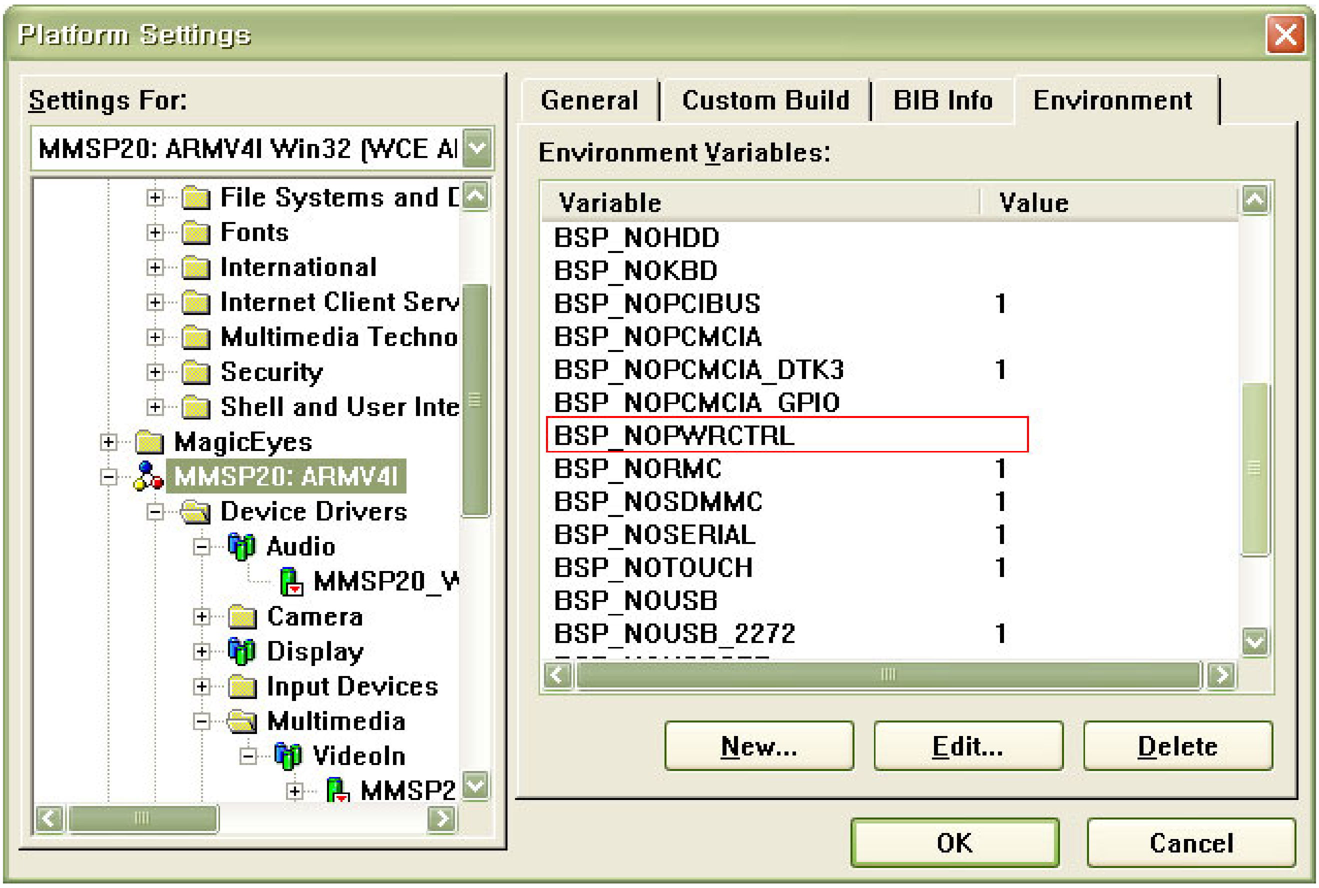
VK_MENU event is not passed to application.

16. Power Manager Driver Guide

16.1 Adding Power Manager Driver

Select **Platform -> Settings** menu and open **Environment** tab.

Add **BSP_NOPWRCTRL** variable and leave the value as **null**



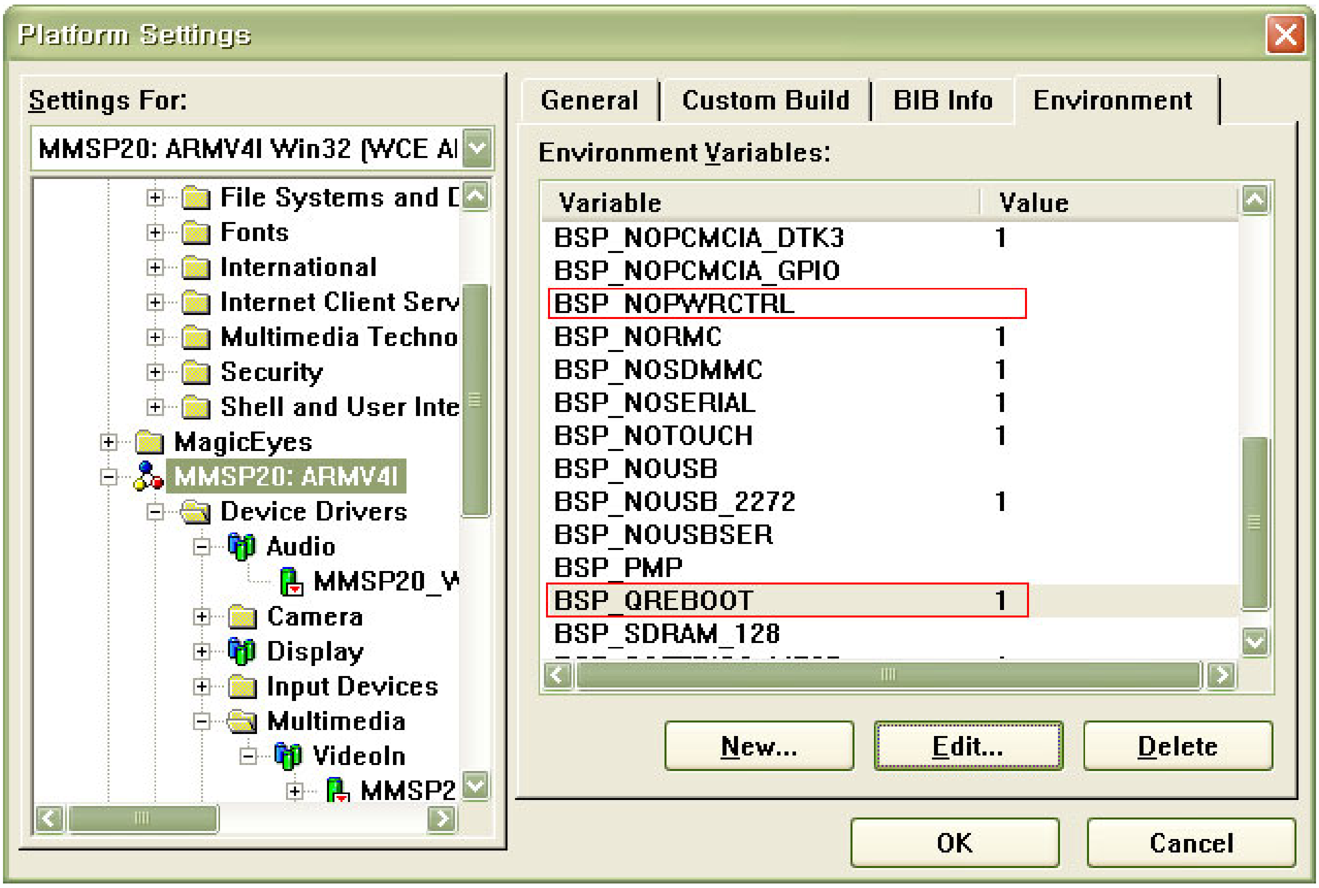
After setting environment variable, build OS image to apply it.

16.2 Using Suspend/Wakeup Function

After setting environment variable and building OS is done, you can use **EXTIN0** and **EXTIN1** button of the DTK 4.11 board for suspend/wakeup. To suspend system, press **EXTIN0** button. To wakeup suspended system, press **EXTIN1** button.

16.3 Quick Booting Mode

You can use quick booting mode *instead of* normal suspend/wakeup mode. Add both **BSP_NOPWRCTRL** and **BSP_QREBOOT** environment variable and set value of these variables as null and **1**.



After setting environment variable, build OS image to apply it.

16.4 Using Quick Booting Mode

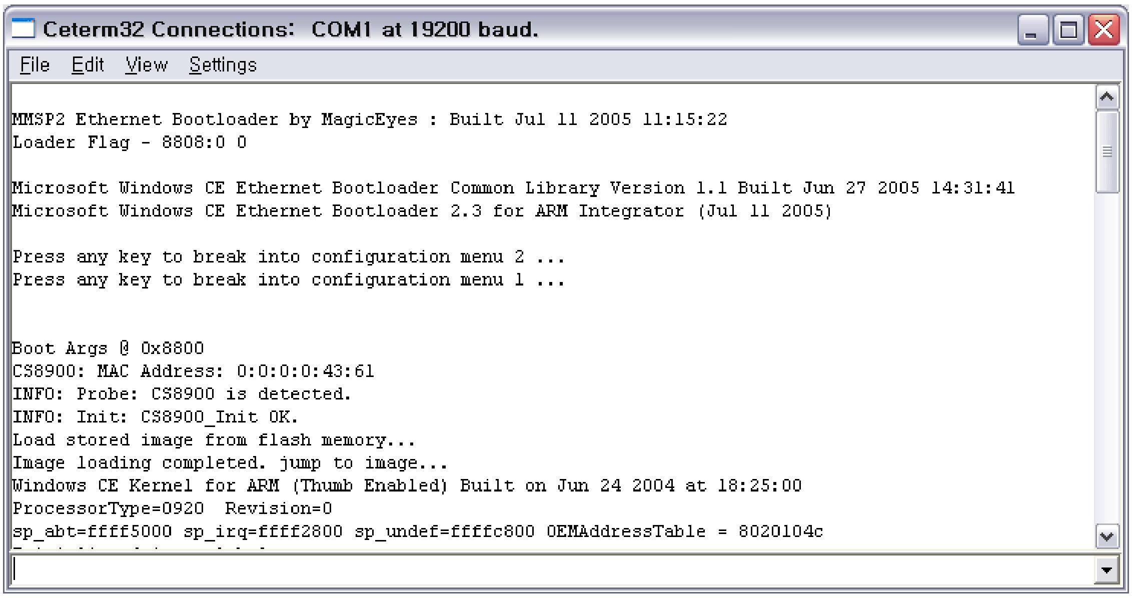
After setting environment variable and building OS is done, you can use **EXTIN0** and **EXTIN1** button of the DTK 4.11 board for quick reboot. To suspend system, press **EXTIN0** button. And when you press **EXTIN1** button at suspend state, system enters into quick boot state.

Quick rebooting is done with this sequence:

- When EXTIN1 button is pressed at suspend state, power manager driver sets quick reboot flag and reset system.
- Ethernet bootloader reads flag.
- If quick reboot flag is set, Ethernet bootloader jumps to OS base address without loading process.

Quick reboot time can be less than 5 seconds if OS does not have drivers or components that take long time for initialization.

This is an example of normal boot. (Boot from image stored in NAND flash)



Ceterm32 Connections: COM1 at 19200 baud.

File Edit View Settings

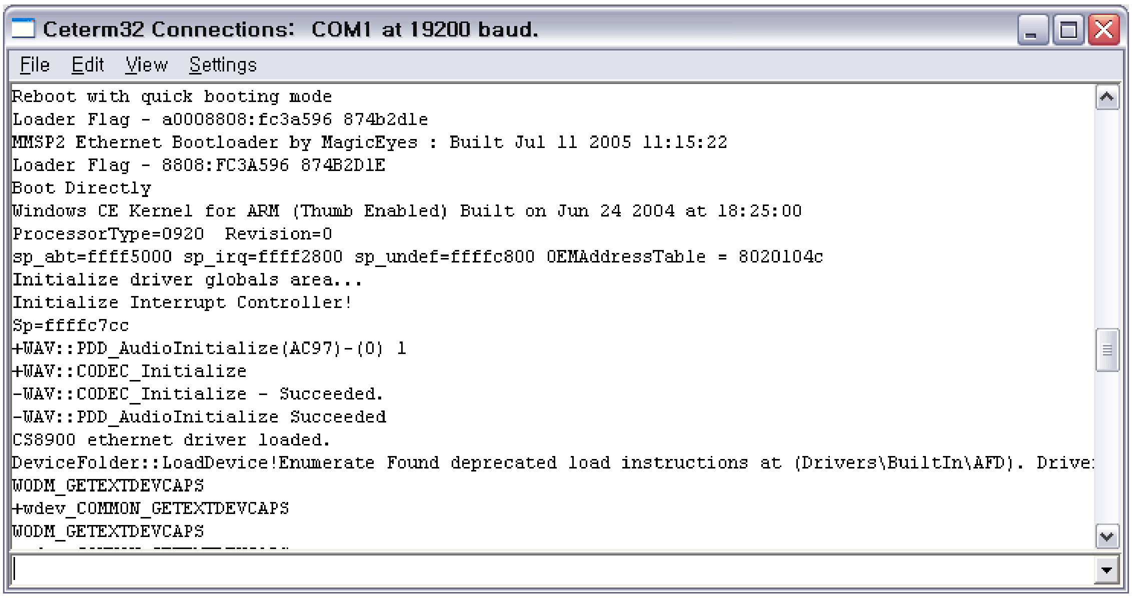
```
MMSP2 Ethernet Bootloader by MagicEyes : Built Jul 11 2005 11:15:22
Loader Flag - 8808:0 0

Microsoft Windows CE Ethernet Bootloader Common Library Version 1.1 Built Jun 27 2005 14:31:41
Microsoft Windows CE Ethernet Bootloader 2.3 for ARM Integrator (Jul 11 2005)

Press any key to break into configuration menu 2 ...
Press any key to break into configuration menu 1 ...


Boot Args @ 0x8800
CS8900: MAC Address: 0:0:0:0:43:61
INFO: Probe: CS8900 is detected.
INFO: Init: CS8900_Init OK.
Load stored image from flash memory...
Image loading completed. jump to image...
Windows CE Kernel for ARM (Thumb Enabled) Built on Jun 24 2004 at 18:25:00
ProcessorType=0920  Revision=0
sp_abt=ffff5000 sp_irq=ffff2800 sp_undef=ffffc800 OEMAddressTable = 8020104c
```

This is an example of quick reboot. (Boot without OS image loading. See loader flag.)



Ceterm32 Connections: COM1 at 19200 baud.

File Edit View Settings

```
Reboot with quick booting mode
Loader Flag - a0008808:fc3a596 874b2d1e
MMSP2 Ethernet Bootloader by MagicEyes : Built Jul 11 2005 11:15:22
Loader Flag - 8808:FC3A596 874B2D1E
Boot Directly
Windows CE Kernel for ARM (Thumb Enabled) Built on Jun 24 2004 at 18:25:00
ProcessorType=0920  Revision=0
sp_abt=ffff5000 sp_irq=ffff2800 sp_undef=ffffc800 OEMAddressTable = 8020104c
Initialize driver globals area...
Initialize Interrupt Controller!
Sp=ffffc7cc
+WAV::PDD_AudioInitialize(AC97)-(0) 1
+WAV::CODEC_Initialize
-WAV::CODEC_Initialize - Succeeded.
-WAV::PDD_AudioInitialize Succeeded
CS8900 ethernet driver loaded.
DeviceFolder::LoadDevice!Enumerate Found deprecated load instructions at (Drivers\BuiltIn\AFD). Drive
WODM_GETEXTDEVCAPS
+wdev_COMMON_GETEXTDEVCAPS
WODM_GETEXTDEVCAPS
```

17. MPEG4HW Decoder Guide

17.1 Specifications

- Supports AVI format only
- Supports MPEG4 Simple Profile and Advanced Simple Profile
- Supports DivX 4.x, DivX 5.x, DivX 3.11 and Xvid

17.2 Performance

- Please see MPEG4 Decoder performance chart.

17.3 Limitations

- Do not support 4MV + QPEL contents.
- Do not support GMC with 2 or more warping points.
- In some DivX 3.11 contents you may see some defections.

17.4 Required Components

See Display Driver Guide to see information about output of decoded result.

Add components shown below in **Core OS** → **Display based devices** →**Multimedia Technologies** → **Multimedia Components** of the catalog.

- **DirectShow** → **DirectShow Core**
- **DirectShow** → **DMO Wrapper Filter**
- **DirectShow** → **Video Codecs and Rednerers** → **Overlay Mixer**
- **DirectShow** → **Video Codecs and Rednerers** → **DirectShow Video Renderer**
- **Windows Media Player** → **Windows Media Player Application**
- **Windows Media Player** → **Windows Media Player OCX**
- **Windows Media Technologies** → **Windows Media Techonologies**

Also Add next three components

- **MMSP20: ARMV4I** → **DSHOW Filters** →**MMSP2 MPEG4 Video Decoder**.
- **MMSP20: ARMV4I** → **DSHOW Filters** →**DST AVI Splitter**
- **MMSP20: ARMV4I** → **DSHOW Filters** →**MES MP3 Decoder Evaluation Version**

18. DST WMV Decoder

18.1 Specifications

- Supports QVGA, 500Kbps
- Supports WMV7, WMV8, WMV9

For more information see DSTWMVDec.doc in MMSP20\DOCS folder.

18.2 Limitations

18.3 Required Components

- All components for MPEG4HW Decoder in **Core OS → Display based devices →Multimedia Technologies**
- **Core OS → Display based devices →Multimedia Technologies →Basic Multimedia →WMV and MPEG-4 Streaming**
- **Third Party →BSPs →MMSP20:ARMV4I →DSHOW Filters →DST WMV DECODER**

19. MES MP3 Decoder

MES MP3 Decoder evaluation version is included. It has 300seconds time-limit.

It supports VBR MP3 in AVI file and gives better performance in video decoding than Microsoft MP3 decoder.

19.1 Required Components

- **Third Party →BSPs →MMSP20:ARMV4I →DSHOW Filters →MES MP3 Decoder Eval.**

20. MESPlayer

MMSP20\APPS\MESPlayer

MESPlayer is a simple media player developed by MagicEyes.Co.Ltd. for application developer. It can play a MPEG4 AVI file and encode camera input. Source code also is included in BSP.

20.1 How to Build

To build MESPlayer, see a document in MMSP20\APPS\MESPlayer folder.

20.2 How to Add into Windows CE image

Copy your MES.exe into MMSP20\FILES folder.

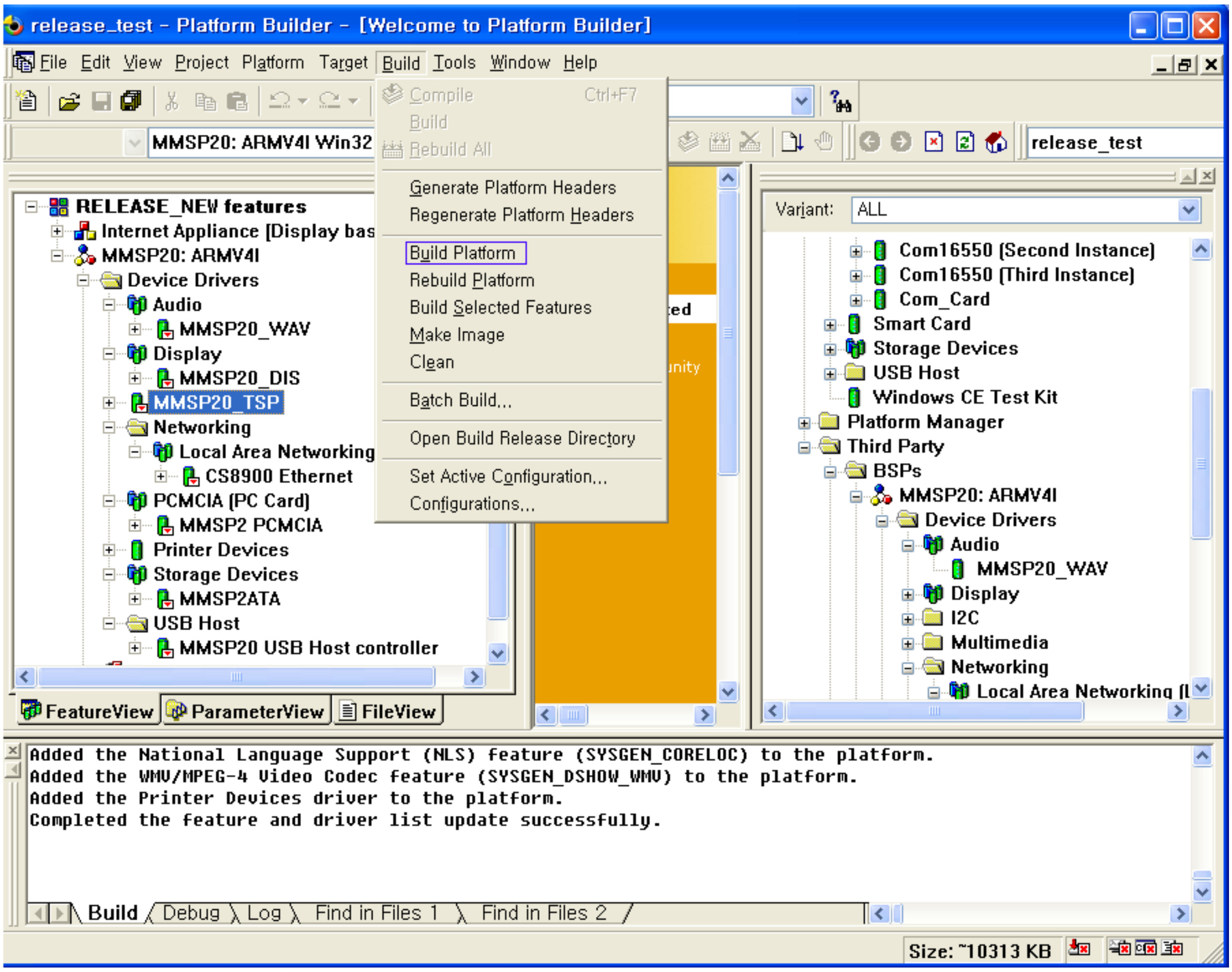
Copy and paste following lines into project.bib.

```
FILES
;   Name              Path                                                Memory Type
;   --------------    ---------------------------------------------    -----------
Mes.exe               $(_WINCEROOT)\PLATFORM\MMSP20\FILES\Mes.exe      NK
```

21. Building WinCE OS

21.1 Building

When setting is completed, start building by selecting **Build** -> **Build Platform**.



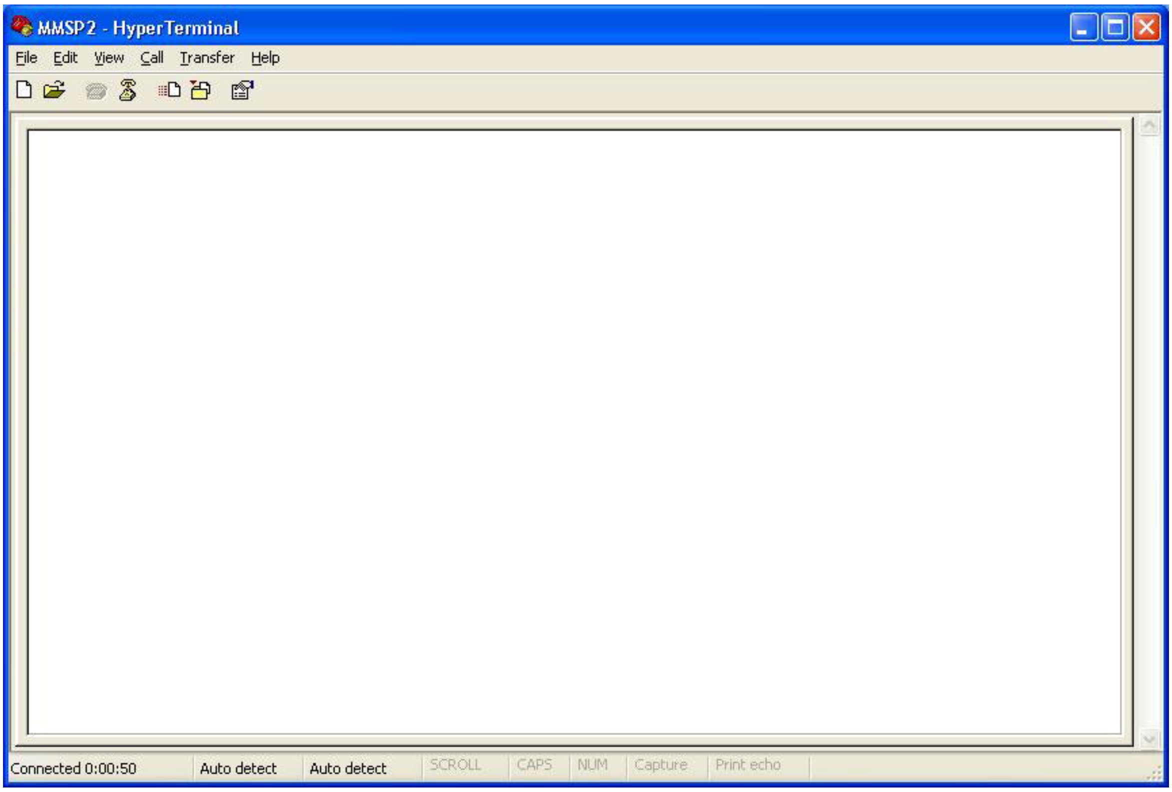
If build is done without error, download new OS image to DTK board. (See next chapter.)

22. Download Image with Ethernet Bootloader

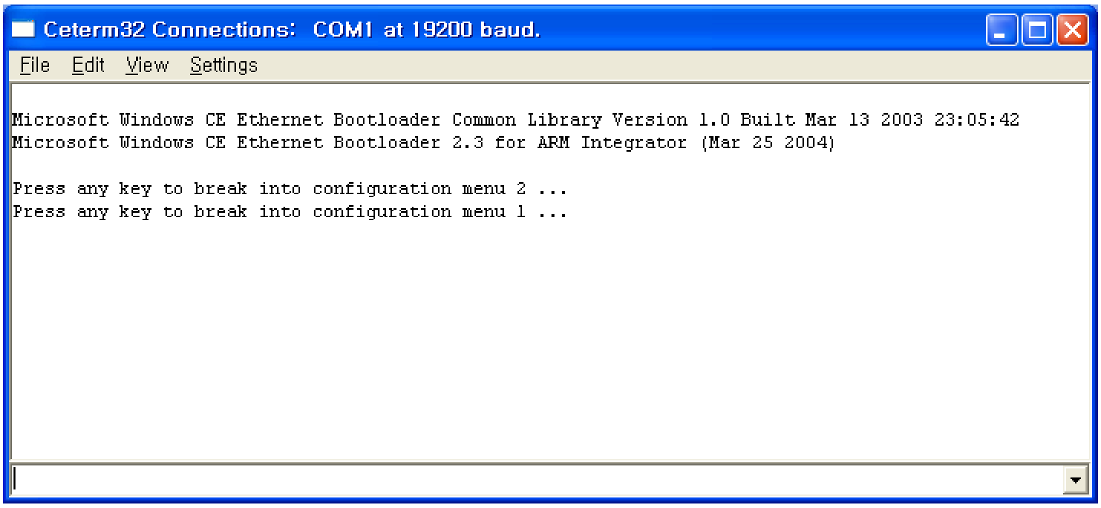
To download Windows CE OS image, make sure ethernet bootloader is correctly installed on DTK board.

22.1 Setting Configuration of Ethernet Bootloader

- Connect PC serial port and debugging serial port of the DTK board with cross cable.
  (Debugging port is right of the two serial ports.)
- Connect LAN Cable on the board.
- Run terminal program on PC and set baud rate to 19200bps.
  EX) START – Programs – Accessories – Communications – HyperTerminal



- When connection and setting is done correctly and power of the DTK board is on, bootloader message is displayed on terminal screen. (Hyper Teminal or Ceterm)



22.2 Settings for OS Image Download

Go to menu by pressing any key when “Press any key to break into configuration menu” message is shown.

22.2.1 Setting Boot Options

Boot options are:

- **Mode1.** Boot with OS image stored in NAND flash memory.
- **Mode2.** Download OS image through Ethernet and store in NAND flash. Boot after writing to flash memory is completed.
- Download OS image through Ethernet and boot without storing to NAND flash.

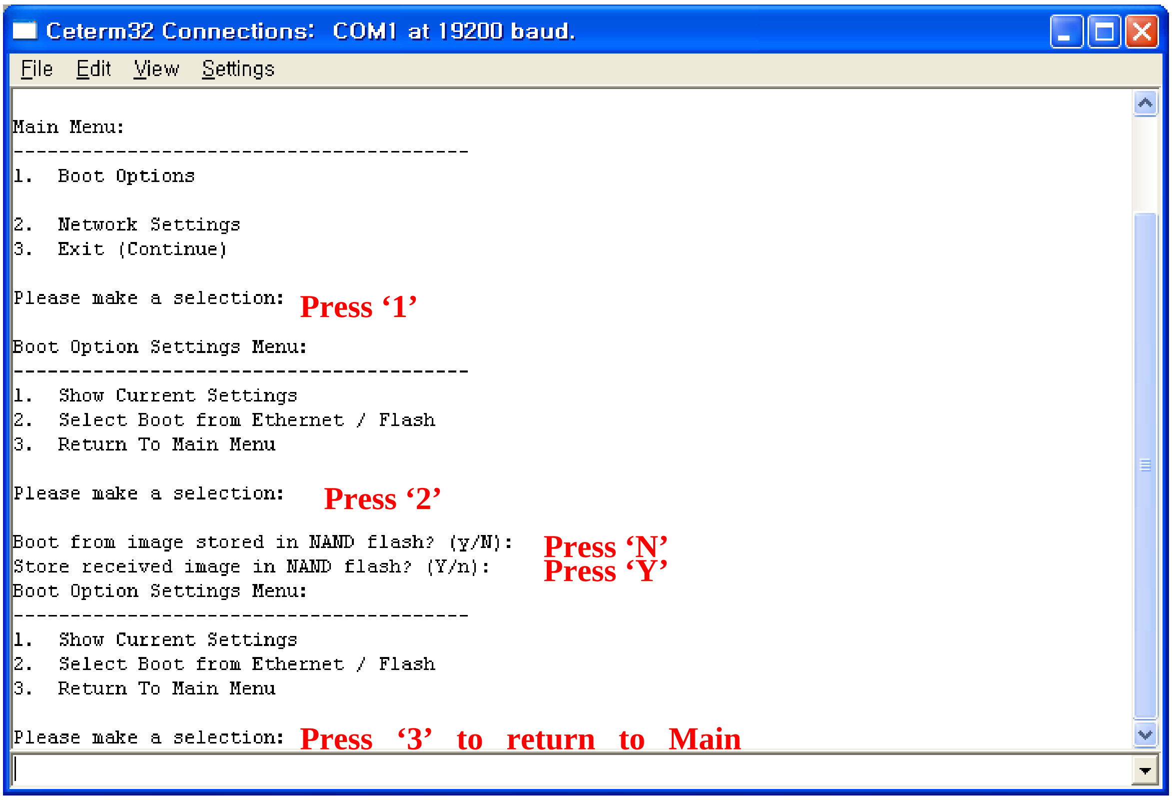


Figure. Mode2-Download OS Image mode

22.2.2 Setting Network Options

Set IP and MAC address for OS image download through Ethernet.

Select **Network Settings** -> **Store Static IP Address and Subnet Mask** and Set IP Address, Subnet mask, MAC address in turn.

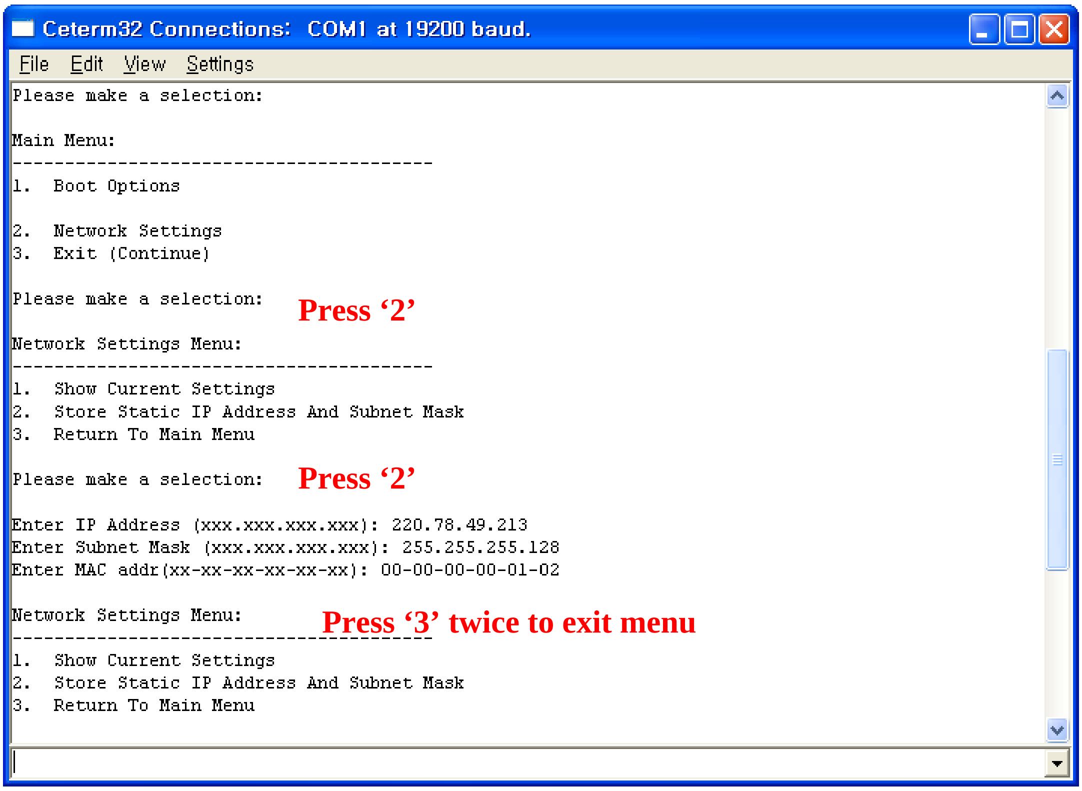


Figure. Network Setting

*Note: The beginning 4 number of the MAC address should be zero. (ex: 00-00-00-00-12-34)

Finishing Configuration Setup

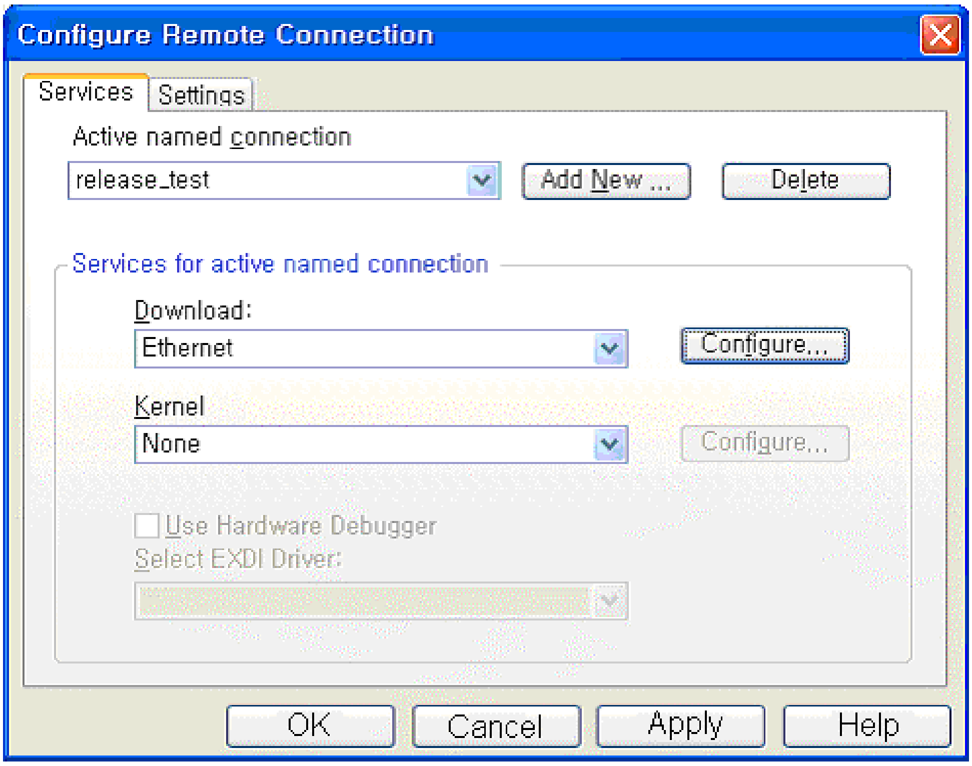
Save options by selecting **Return to Main Menu** -> **Exit**.

Now bootloader will be run according to the stored options.

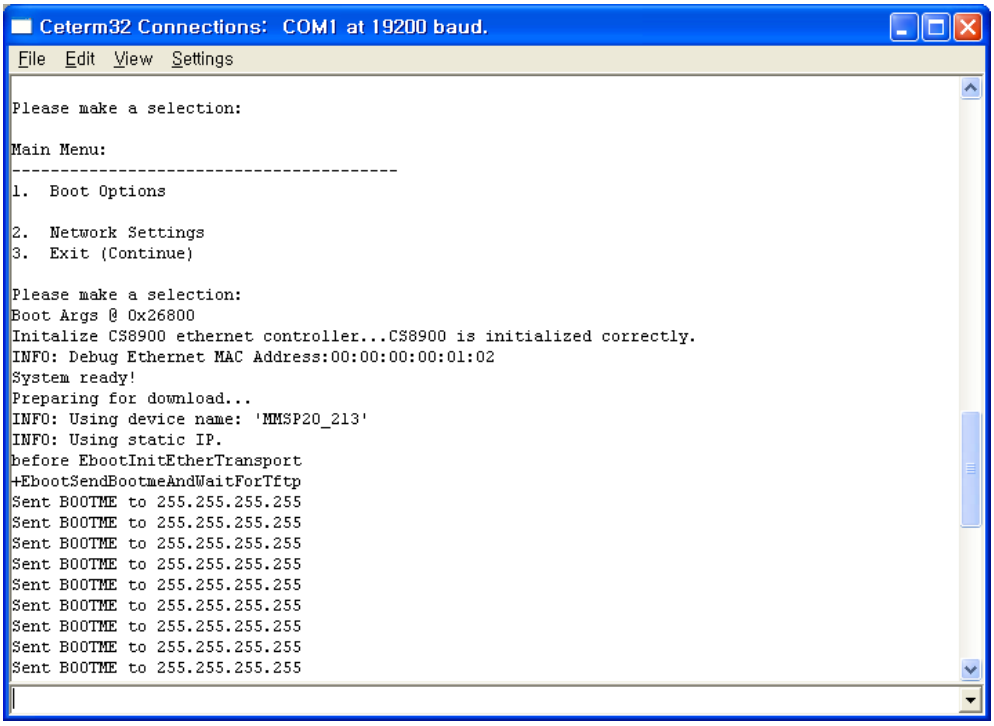
Setting Platform Builder for OS Image Download

Select **Target** -> **Configure Remote Connection** in menu.

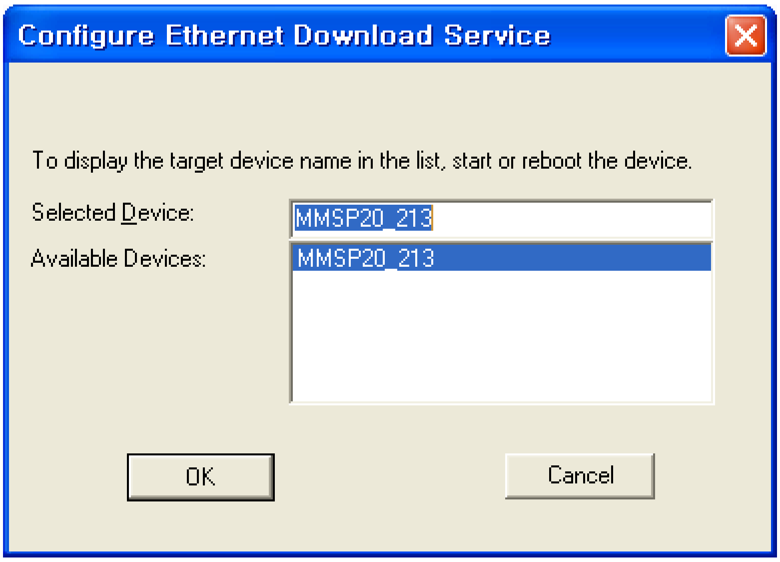
Set **Download** as 'Ethernet' and prepare to select download target by pressing 'Configure' button.



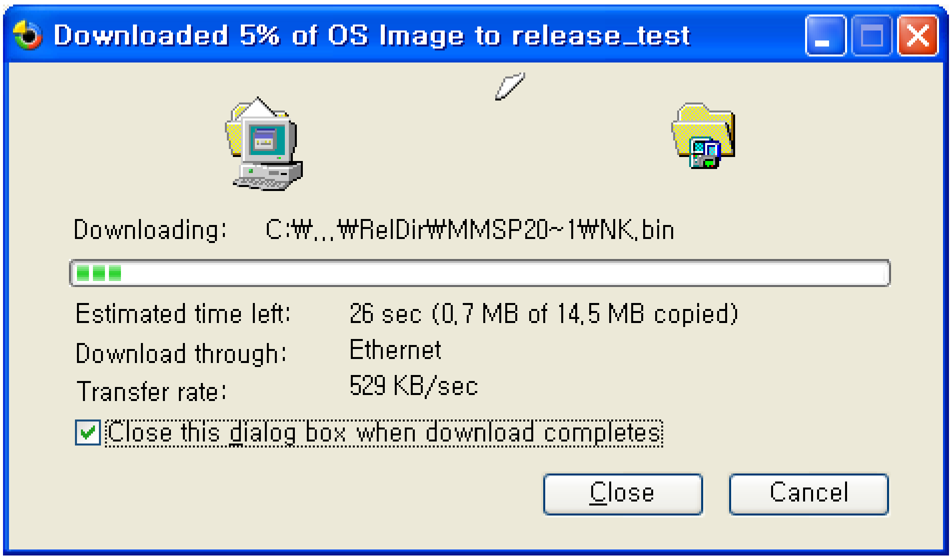
Bootloader sends BOOTME packet when download mode is selected as Ethernet download.



Select target device name when it is shown in **Available Devices** list.



Finish configuration and run Ethernet download by selecting **Target** -> **Download/Initialize**. (Starting download is possible only when bootloader sends BOOTME packet.)



After downloading is completed, OS image runs. If you selected option that store received image to flash memory, it takes some time to store image and run OS.

22.2.3 NAND Flash mode Setting

After store OS Image, User have to set Mode 1. to boot NAND Flash mode

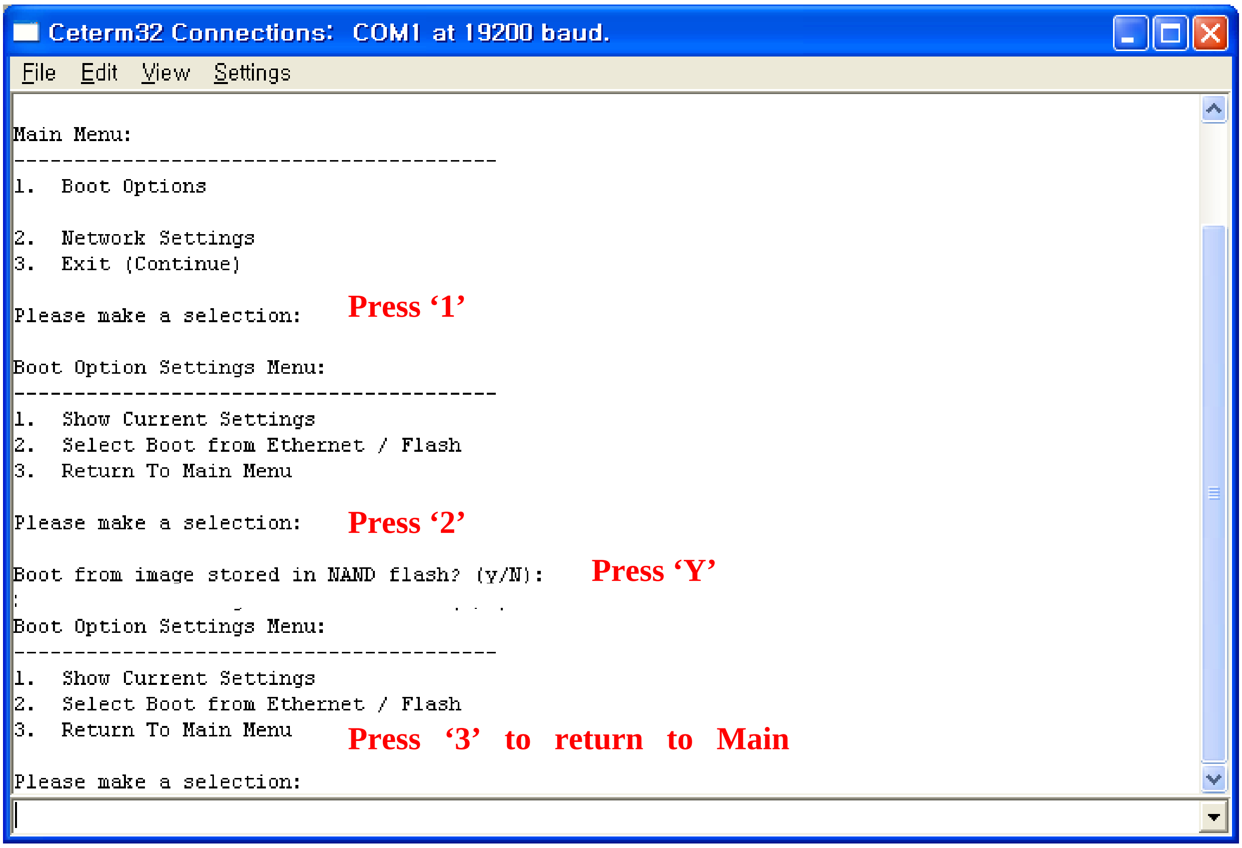


Figure. Mode1-NAND Flash mode

23. Installing Bootloader Image to NAND Flash

Bootloader for Windows CE image is installed when DTK board is delivered. You can replace this bootloader image with ARM debugging tools and DTK diagnostic program.

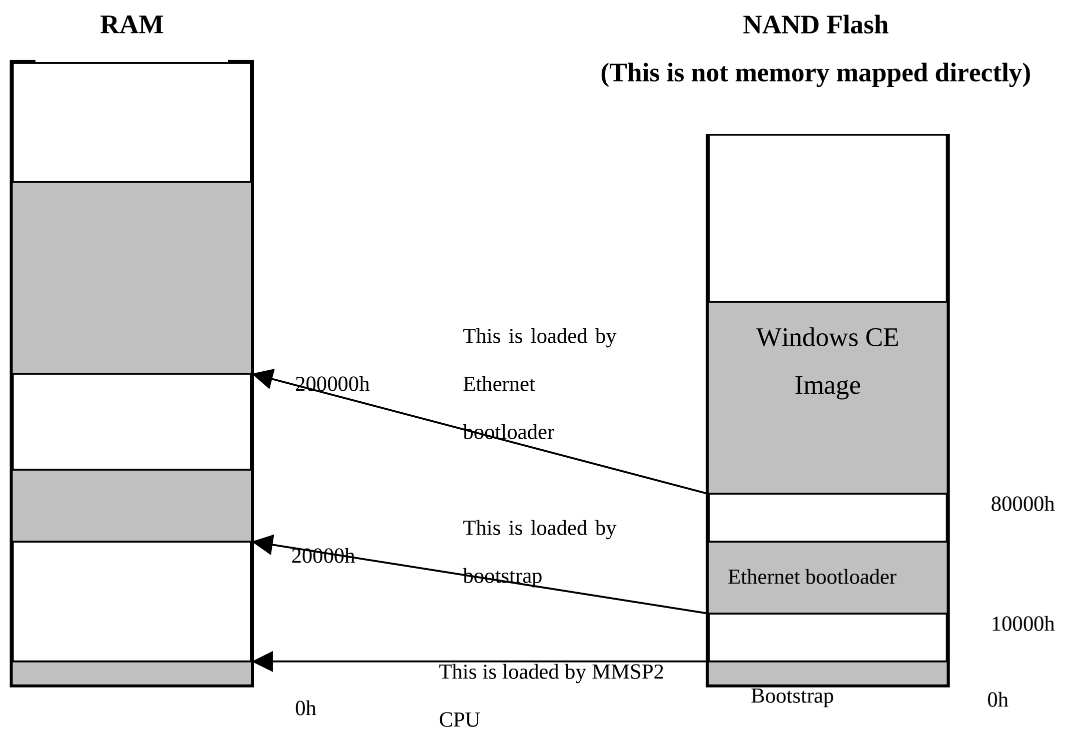
23.1 Boot Sequence

Current version of bootloader used on MMSP2 DTK board is divided with two steps of images. First bootlaoder is a small image (less than 512 bytes) and called 'bootstrap.' This bootstrap is loaded by MMSP2 CPU and it launches second bootloader. This second bootloader is Ethernet bootloader that you have seen in previous chapter.

E. When power is on, MMSP2 CPU loads bootstrap stored in address 0h of NAND flash. This bootstrap image is located in address 0h of the RAM.

F. As bootstrap runs, it loads Ethernet bootloader image stored in 10000h of NAND flash. This Ethernet bootloader image is located in address 20000h of the RAM.

G. Ethernet bootloader loads Windows CE image from NAND flash or get new image through Ethernet.



23.2 Installing Bootloader Image

Because address of NAND flash is not memory mapped directly, a program that accesses to NAND flash interface is needed to write images in NAND flash.

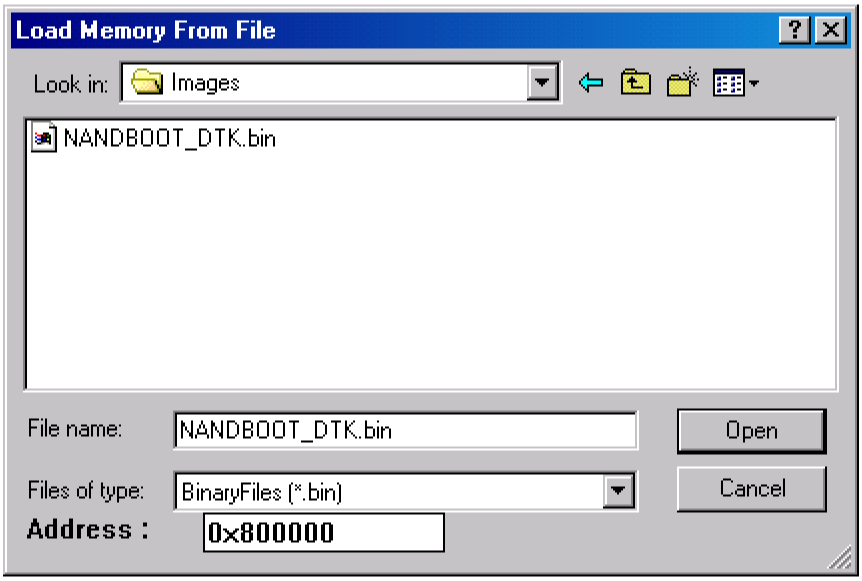
23.2.1 Requirements

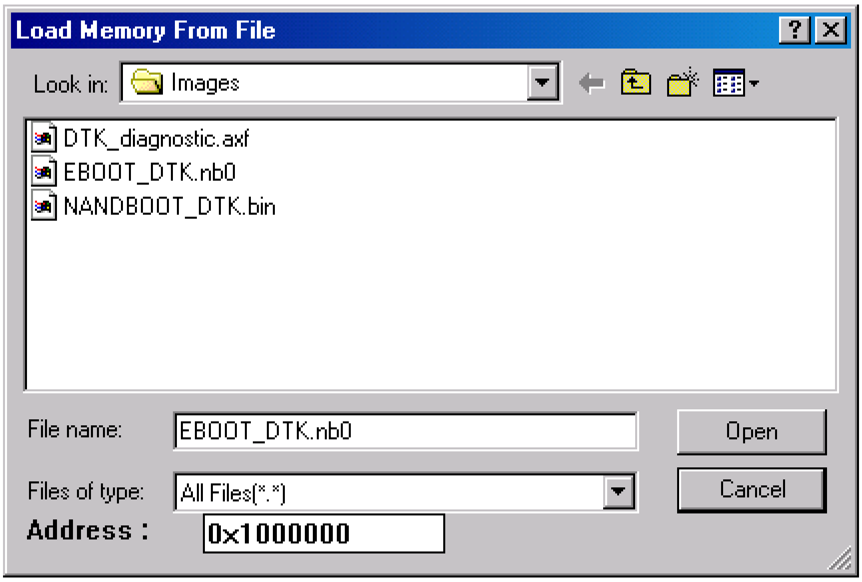
To install or replace bootloader image, you must have:

- A JTAG emulator (OPENice32-A900 or OPENice-A1000)
- A debugger (Spider)
- MMSP2 DTK diagnostic program image

23.2.2 Installing bootloader images

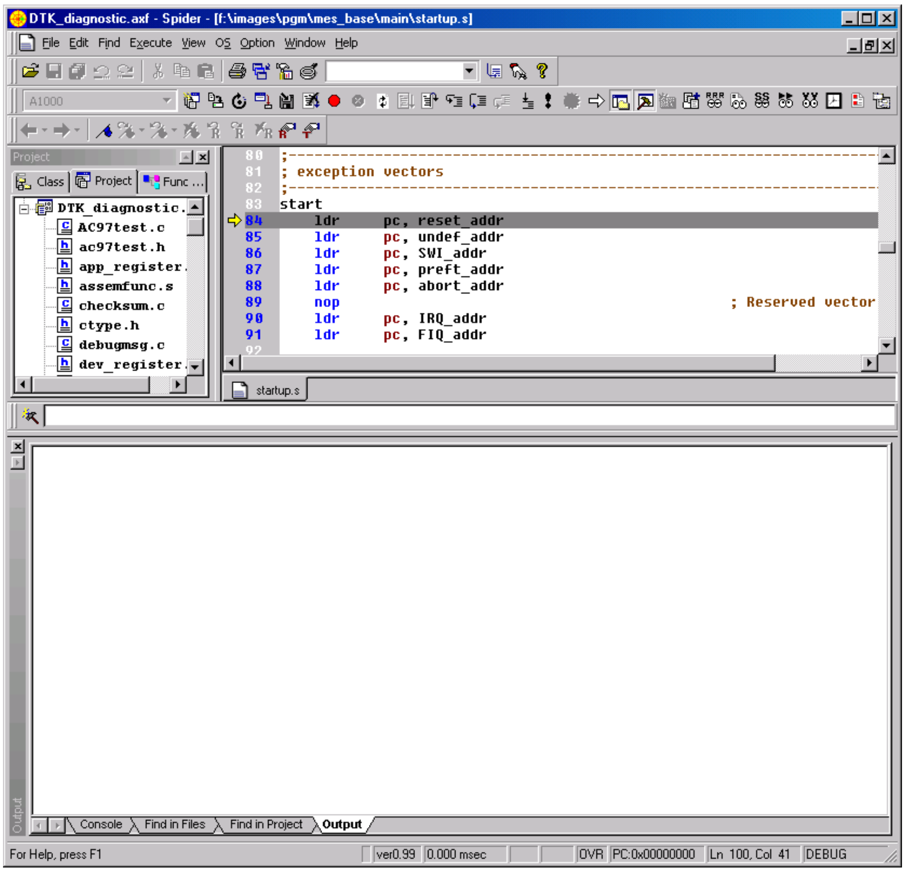
A. Run Spider.
B. Select **File** -> **Load Memory From File** to load new bootloader image to RAM. Address can be any address of RAM area that does not have any interference with DTK diagnostic program. In this guide, we load NANDBOOT_DTK.bin (bootstrap image) to 800000h and EBOOT_DTK.bin (Ethernet bootloader image) to 1000000h.



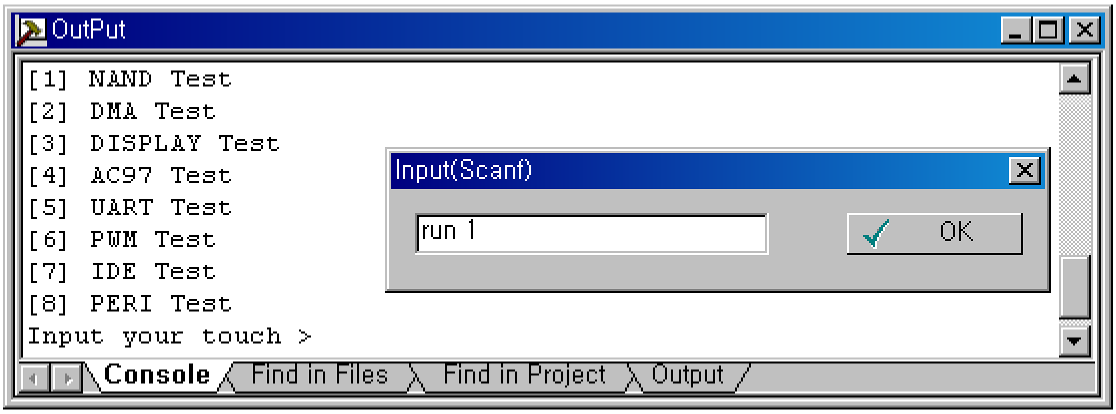


C. Select **File -> Load Image** and load DTK diagnostic program image. (**DTK_diagnostic.axf** or so)

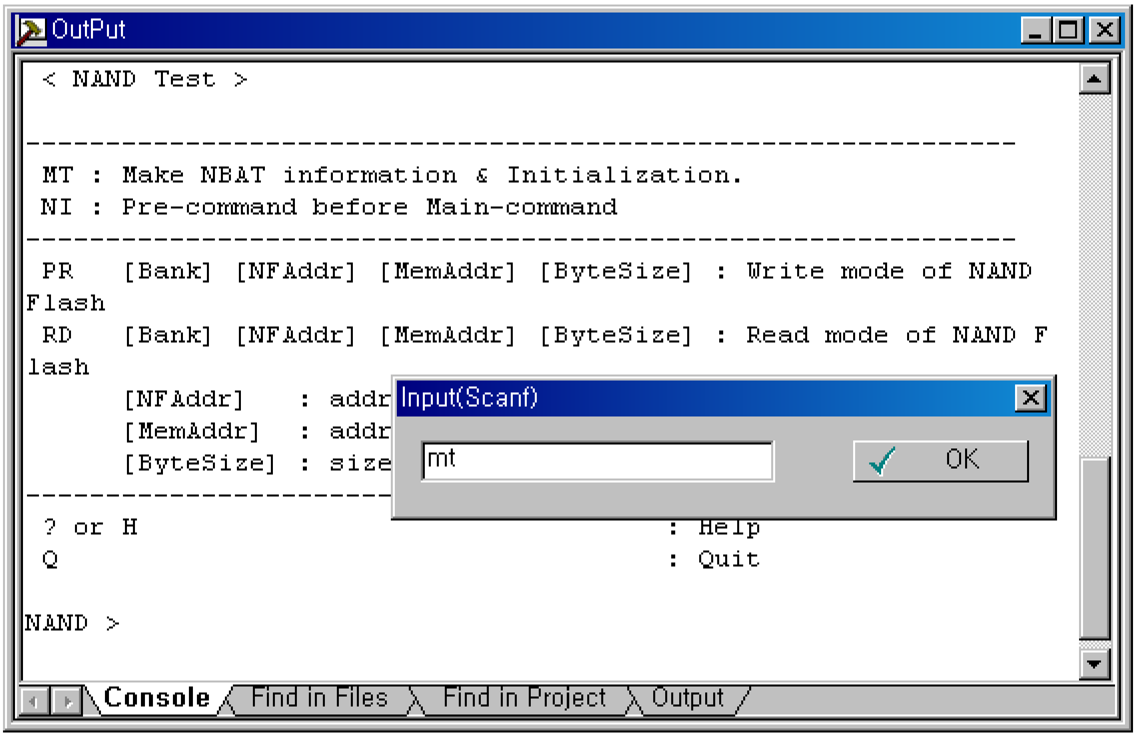
D. Select **Execute -> Go** or Press F5 to run diagnostic program. (You may do this twice to see menu displayed.)

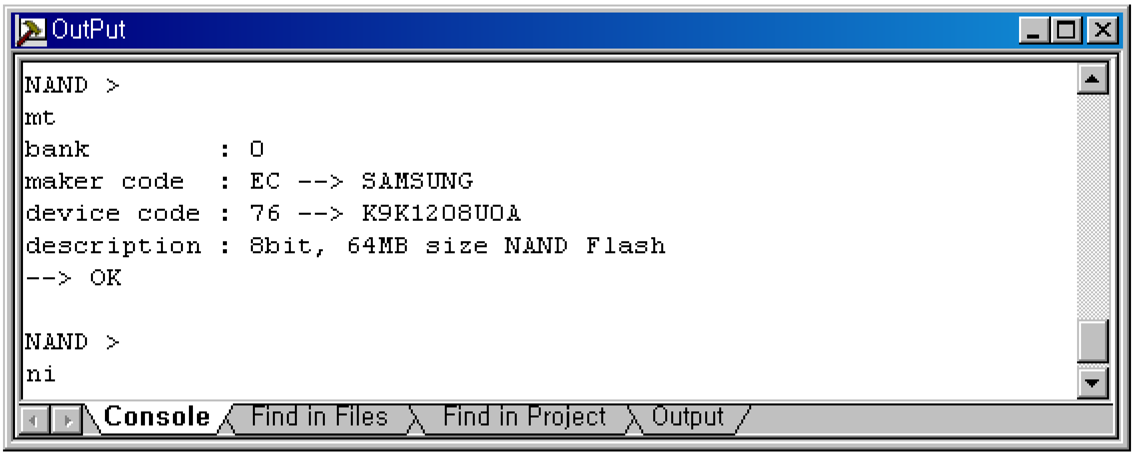


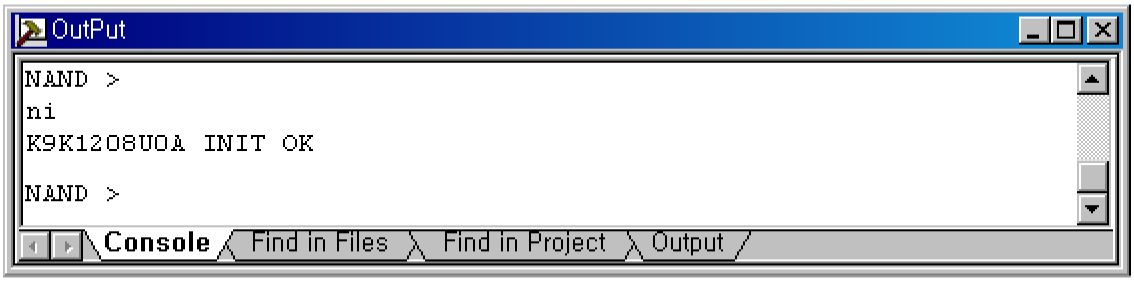
E. Execute "run 1" command in the console window to invoke the "NAND test" mode.



F. Execute "MT" and "NI" command







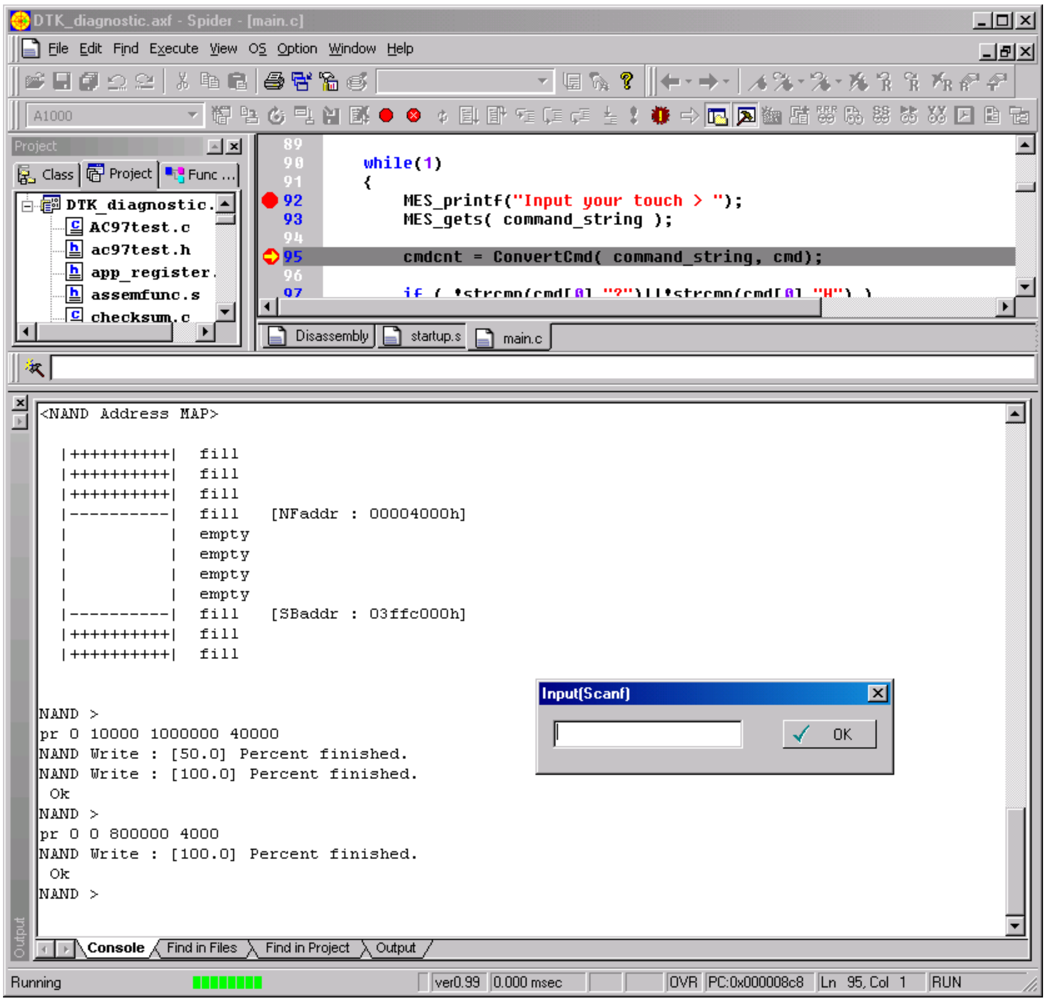
G. Now execute "PR" command to write bootloader images loaded on RAM.

To write Ethernet bootloader image loaded in 1000000h of the RAM, run command like this:

pr 0 10000 1000000 40000

To write bootstrap image loaded in 800000h of the RAM, run command like this:

pr 0 0 800000 4000


DTK_diagnostic.axf - Spider - [main.c]
File Edit Find Execute View OS Option Window Help
while(1)
{
MES_printf("Input your touch > ");
MES_gets(command_string);
cmdcnt = ConvertCmd(command_string, cmd);
Disassembly startup.s main.c
<NAND Address MAP>
|++++++++++| fill
|++++++++++| fill
|++++++++++| fill
|----------| fill [NFaddr : 00004000h]
| | empty
| | empty
| | empty
| | empty
|----------| fill [SBaddr : 03ffc000h]
|++++++++++| fill
|++++++++++| fill
NAND >
pr 0 10000 1000000 40000
NAND Write : [50.0] Percent finished.
NAND Write : [100.0] Percent finished.
Ok
NAND >
pr 0 0 800000 4000
NAND Write : [100.0] Percent finished.
Ok
NAND >
Input(Scanf)
OK
Console Find in Files Find in Project Output
Running ver0.99 0.000 msec OVR PC:0x000008c8 Ln 95, Col 1 RUN

24. Using Debugger with KITL Connection

KITL (Kernel Independent Transport Layer) is a communication layer that manages communication between the development workstation and the target device. You can use KITL for kernel debugging and platform debugging. This BSP supports KITL link with CS8900 Ethernet controller.

Note:

- You should build OS with **Debug** mode for source level debugging.
- Size of Debug mode OS image is much bigger than image with Release mode. Make sure OS image size is less than 30MB. To reduce OS image size, remove large size components such as internet explorer.
- You cannot use KITL with Ethernet and CS8900 Network driver simultaneously. Make sure CS8900 Network driver is not added from catalog.

24.1 Building CS8900 Ethernet Debugging Library

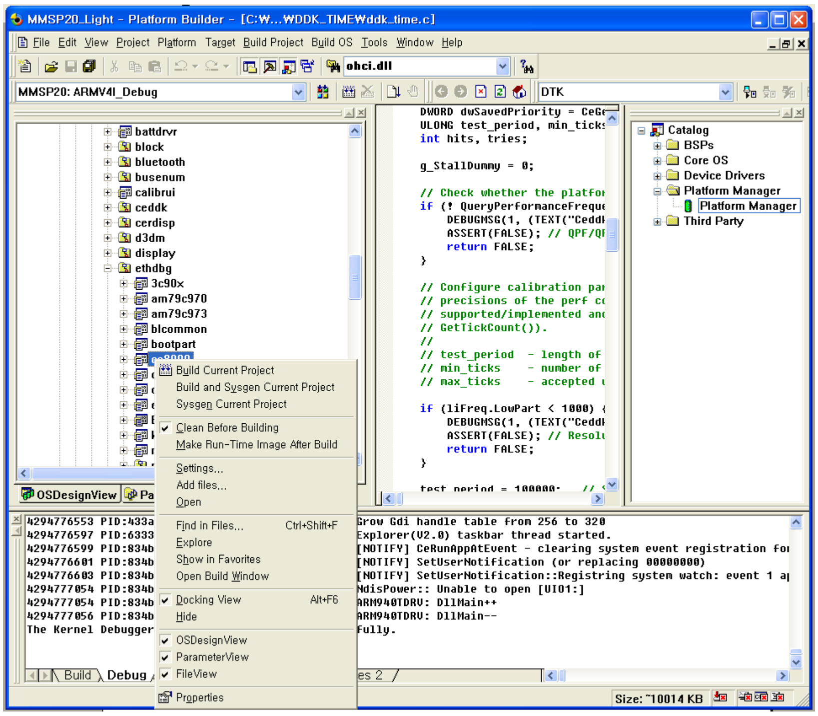
You must modify a source of CS8900 Ethernet debugging library. Location of the source file of this library is **\WINCE420\PUBLIC\COMMON\OAK\DRIVERS\ETHDBG\CS8900**.

Open **cs8900dbg.c** file.

Find the line shown below. Remove this line or cover it with a comment

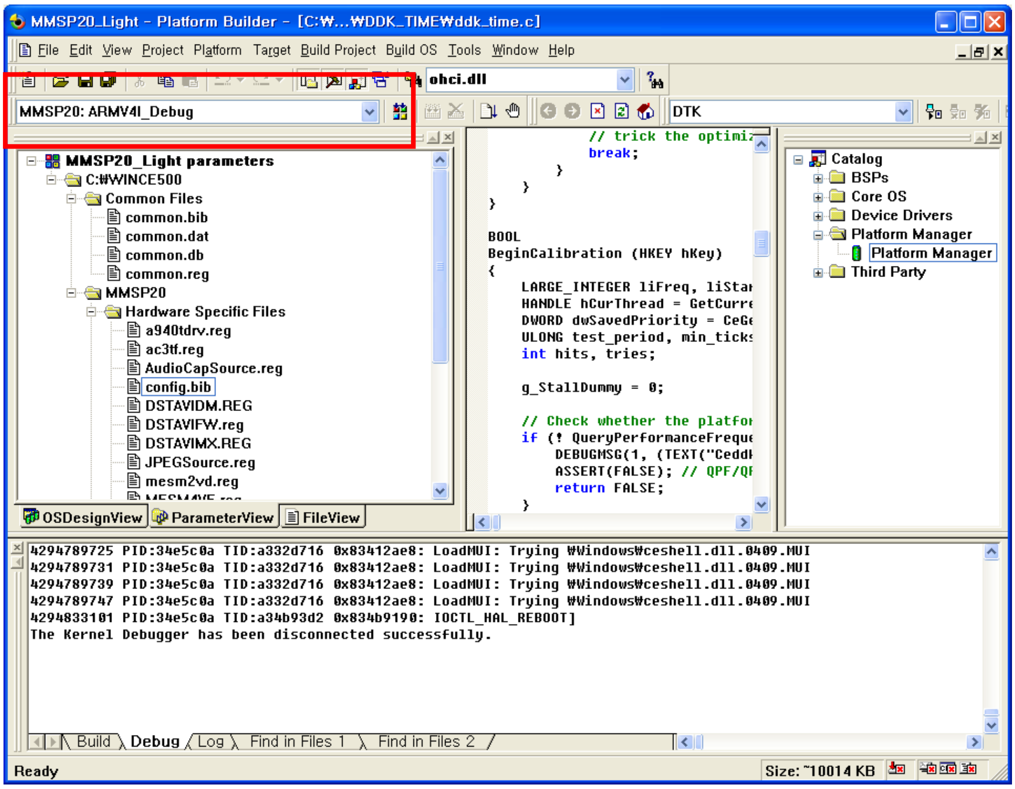
```
#define CS8900_MEM_MODE
```

Build this library. Select CS8900 directory on **FileView** and click it with right button of the mouse. Then perform building by selecting **Build Current Project**. (For detailed information of building part of source code, see **Building WinCE OS** -> **Building Part of BSP Source** chapter of this document.)



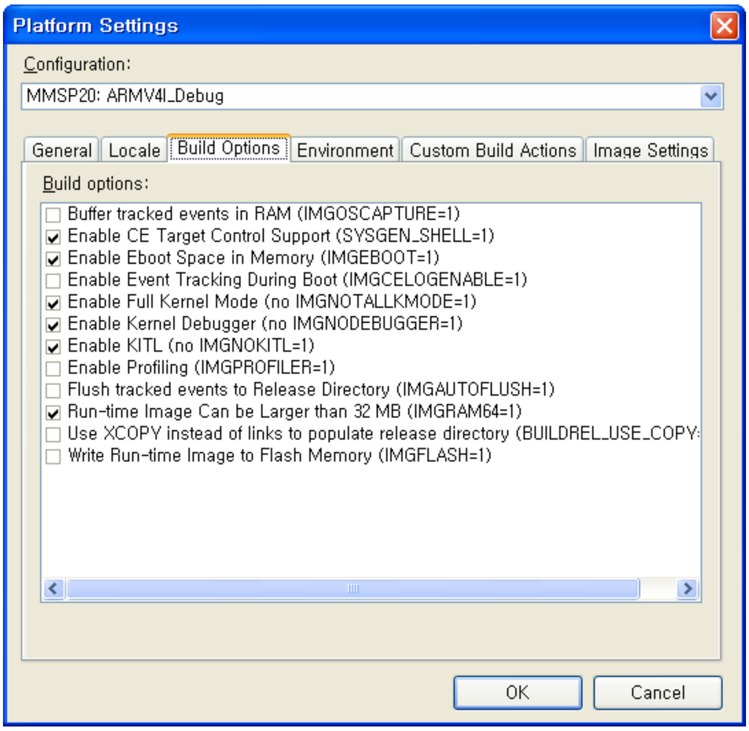
24.2 Setting and Building OS Image for KITL Link

For source level debugging, OS image should be built Debug mode.

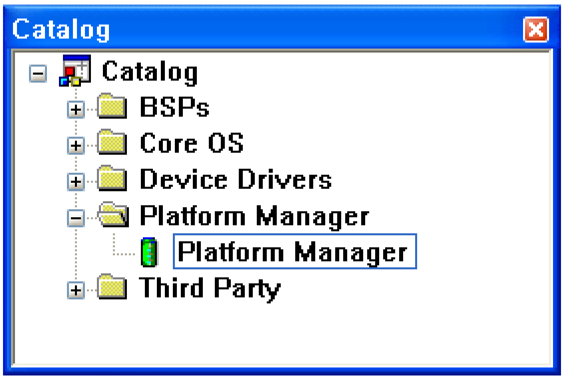


Select **Platform** -> **Settings** menu and select **Build Options** tab.

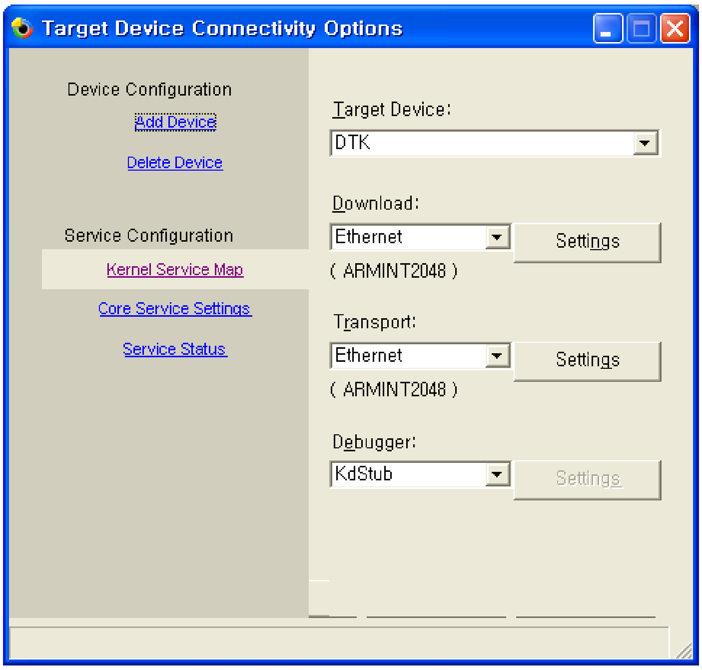
Select **Enable Kernel Debugger** and **Enable KITL** options.



Add **Platform Manager** -> **Platform Manager** from **Catalog** list.



Select **Target** -> **Connectivity Options** and select **Debugger** of current connectivity selection as **Ethernet**.

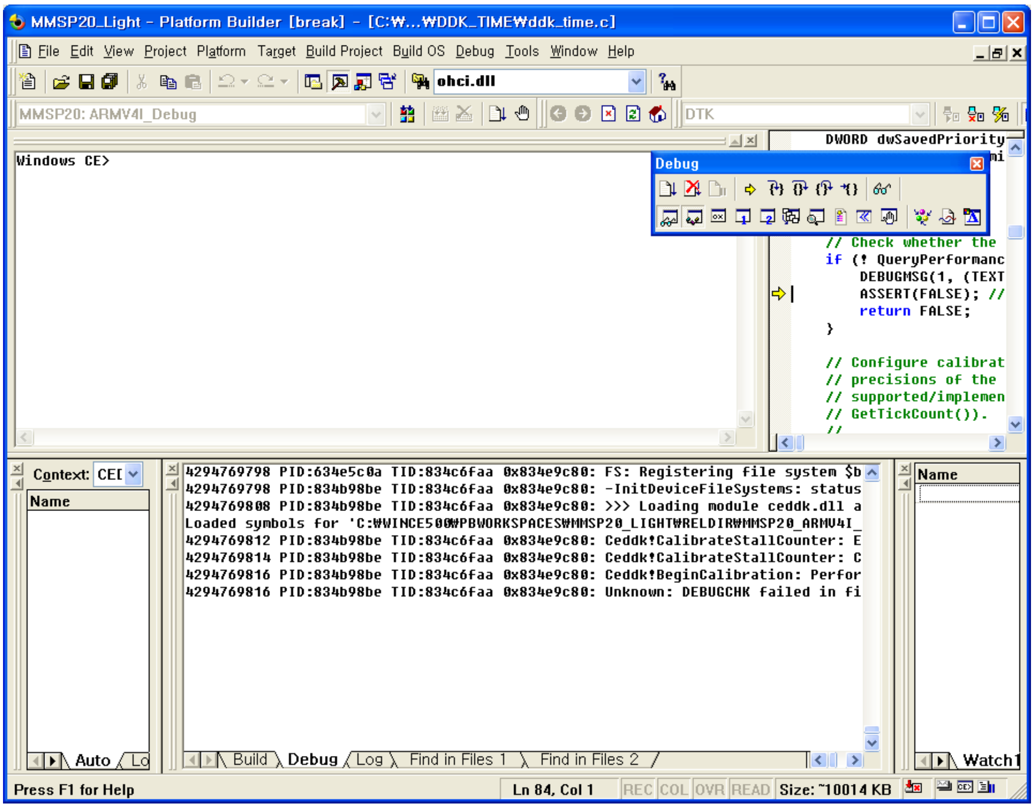


Build OS by selecting **Build OS** -> **Sysgen**.

24.3 Connect Platform Builder and Target System with KITL

Download OS image with Ethernet bootloader.

After download is completed, you can see this debugging window. Now you can use kernel debugging.


MMSP20_Light - Platform Builder [break] - [C:₩...₩DDK_TIME₩ddk_time.c]
File Edit View Project Platform Target Build Project Build OS Debug Tools Window Help
ohci.dll
MMSP20: ARMV4I_Debug
DTK
Windows CE>
Debug
DWORD dwSavedPriority
// Check whether the
if (! QueryPerformanc
DEBUGMSG(1, (TEXT
ASSERT(FALSE); //
return FALSE;
}
// Configure calibrat
// precisions of the
// supported/implemen
// GetTickCount()).
Context: CEI
Name
4294769798 PID:634e5c0a TID:834c6faa 0x834e9c80: FS: Registering file system $b
4294769798 PID:834b98be TID:834c6faa 0x834e9c80: -InitDeviceFileSystems: status
4294769808 PID:834b98be TID:834c6faa 0x834e9c80: >>> Loading module ceddk.dll a
Loaded symbols for 'C:₩WINCE500₩PBWORKSPACES₩MMSP20_LIGHT₩RELDIR₩MMSP20_ARMV4I_
4294769812 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!CalibrateStallCounter: E
4294769814 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!CalibrateStallCounter: C
4294769816 PID:834b98be TID:834c6faa 0x834e9c80: Ceddk!BeginCalibration: Perfor
4294769816 PID:834b98be TID:834c6faa 0x834e9c80: Unknown: DEBUGCHK failed in fi
Auto
Build
Debug
Log
Find in Files 1
Find in Files 2
Watch1
Press F1 for Help
Ln 84, Col 1
REC COL OVR READ
Size: ~10014 KB

25. Using Debugging Tools with TCP/IP Networking

Debugging application developed in eMbedded Visual C++ or using other remote debugging tools are possible with TCP/IP transport connection to DTK board.

Available remote tools are:

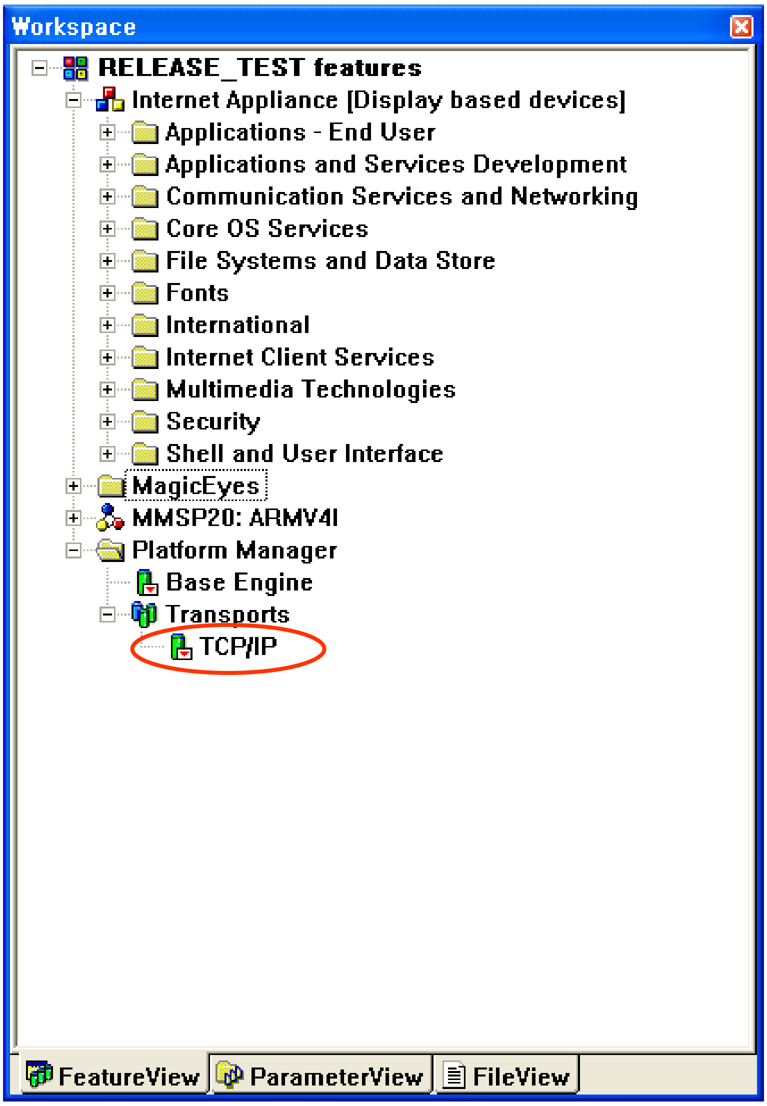
- Remote Call Profiler
- Remote File Viewer
- Remote Heap Walker
- Remote Kernel Tracker
- Remote Performance Monitor
- Remote Process Viewer
- Remote Registry Editor
- Remote SPY
- Remote System Information
- Remote Zoom-in

Local Area Network driver should be set correctly to use these tools.

25.1 Required Components

Add components needed for connection to target board.

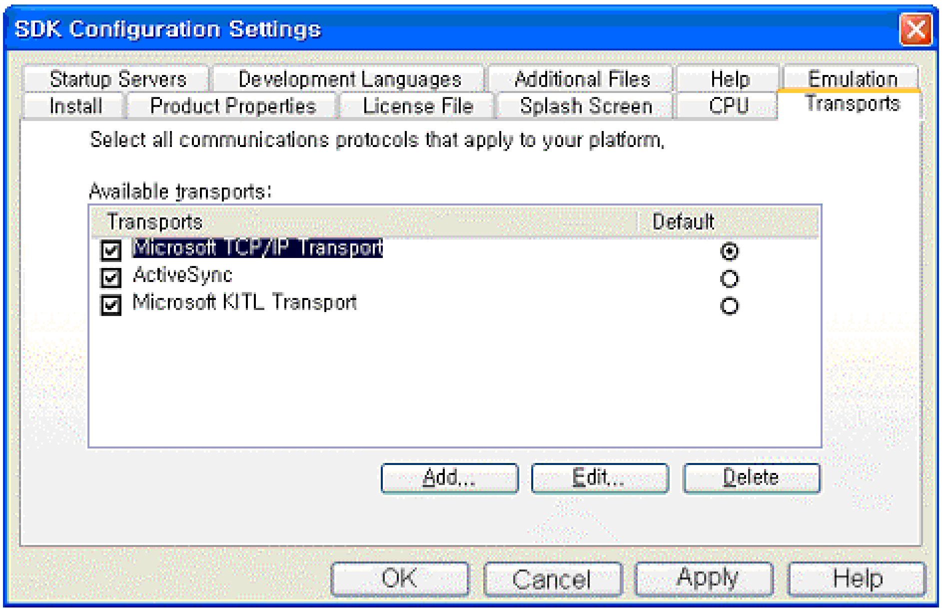
- Platform Manager → Base Engine
- Platform Manager → Transports → TCP/IP

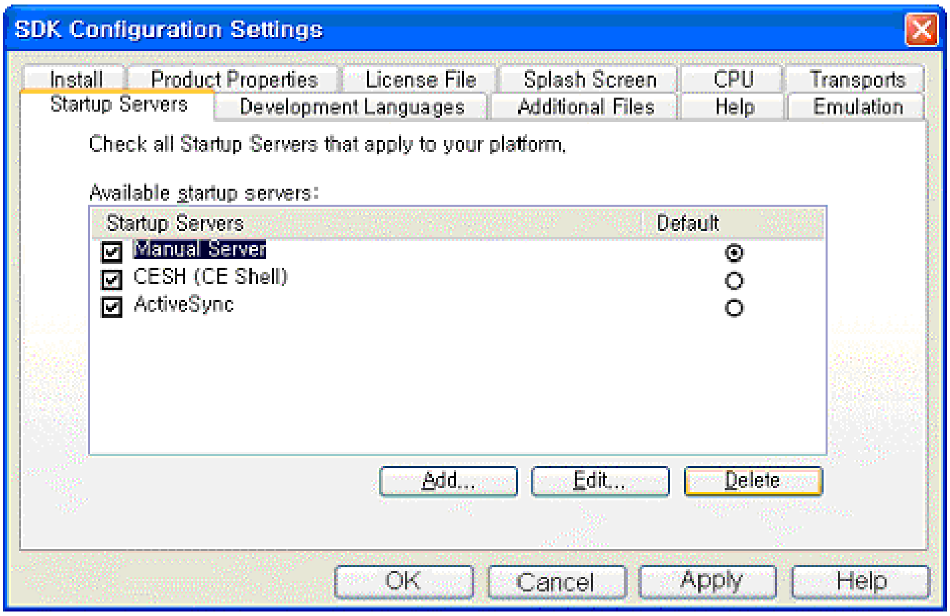


After add all required components, you must rebuild platform to apply new setting.

25.2 Building and Installing SDK for Target Connection

Select **Platform** → **Configure SDK** menu and set configuration as shown below in wizard.





After setting is completed, select Platform → Build SDK and build SDK.

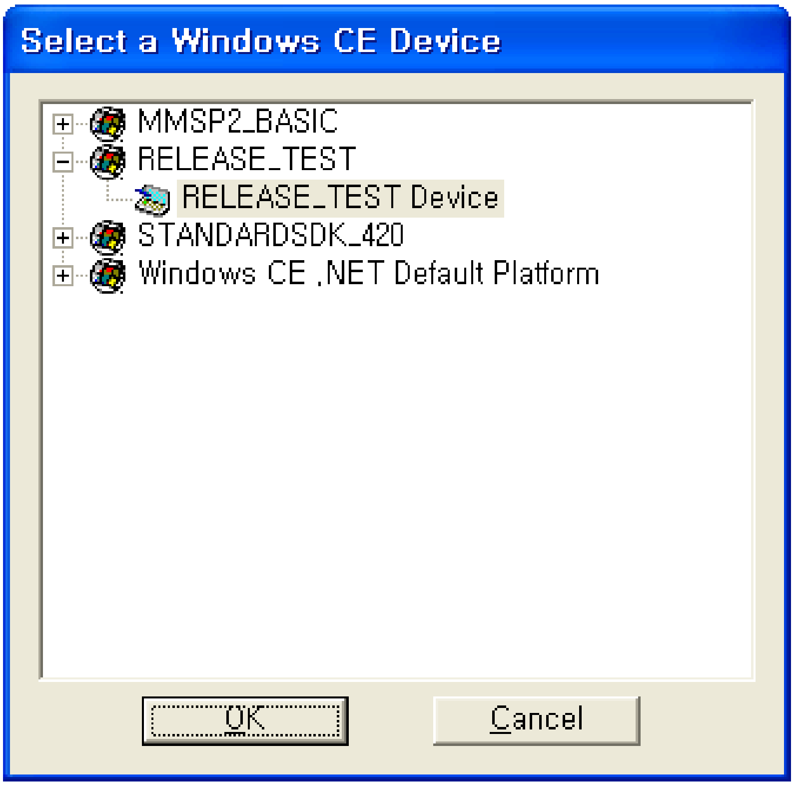
Go to folder that has built SDK and run .msi to install SDK.

Folder is WINCE420\PUBLIC\[name of platform]\SDK

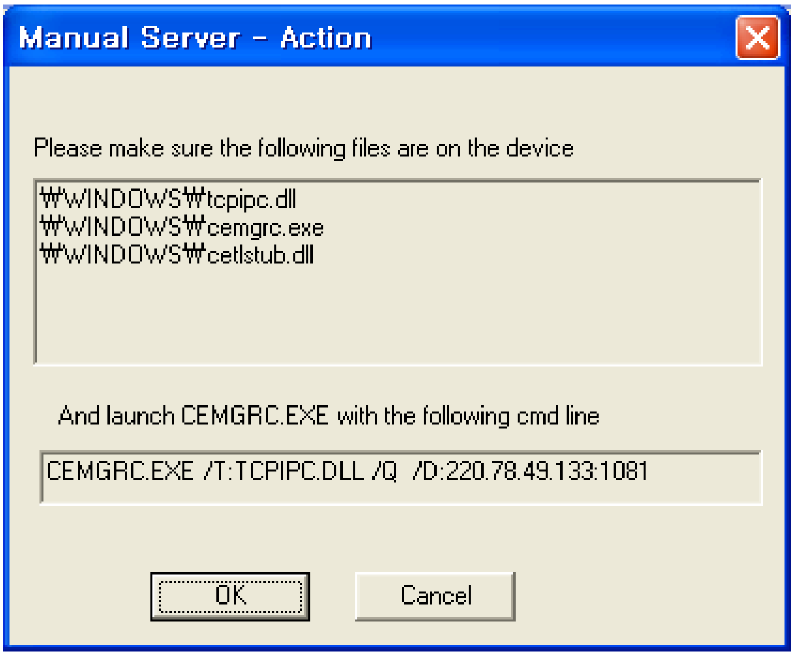
* Reboot system to apply installed SDK.

25.3 Connecting to Target by Using Debugging Tools

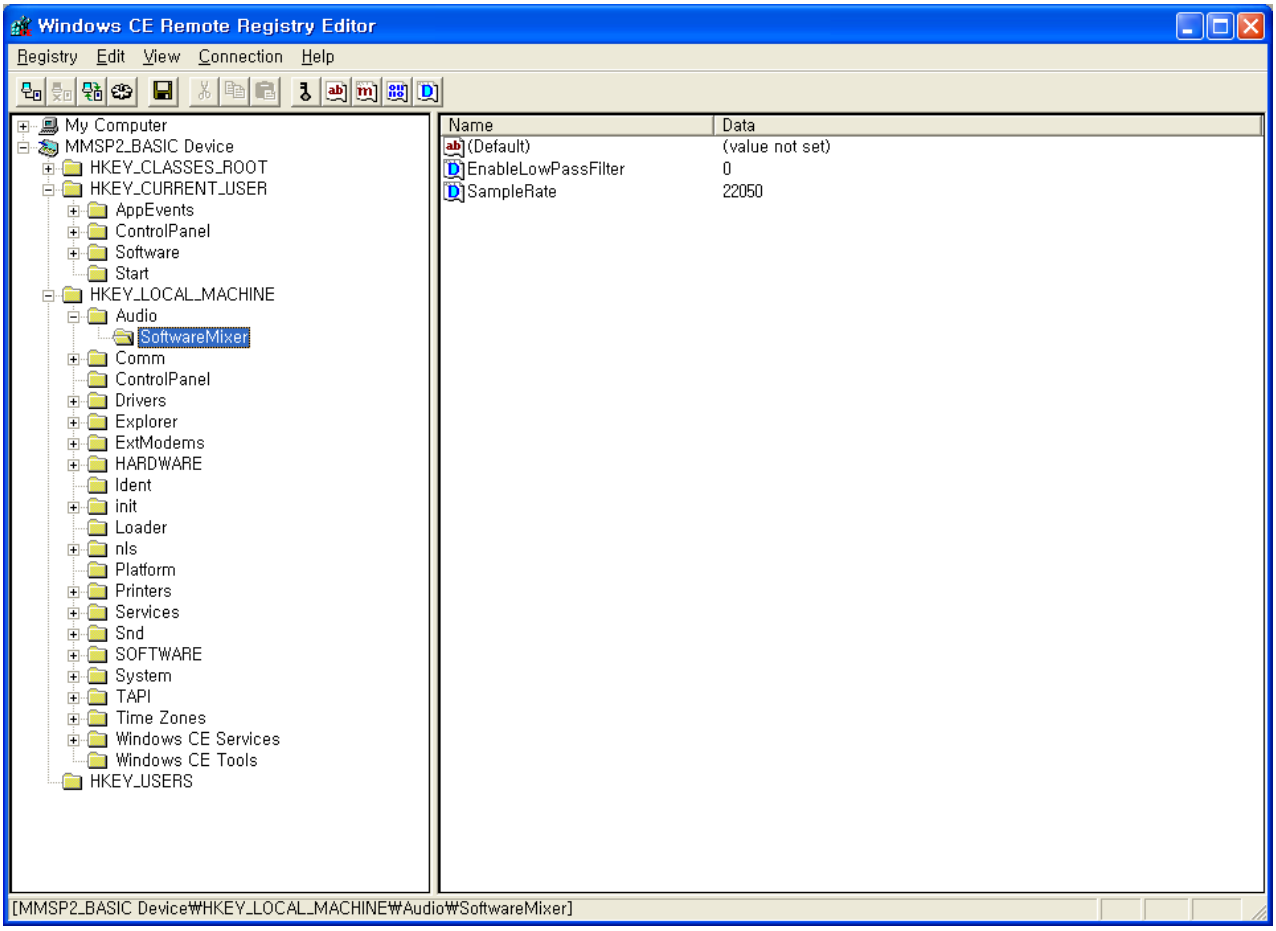
Select device of newly installed SDK when selection window is shown.



Select Start -> Run by pressing start button in DTK board and run as shown.

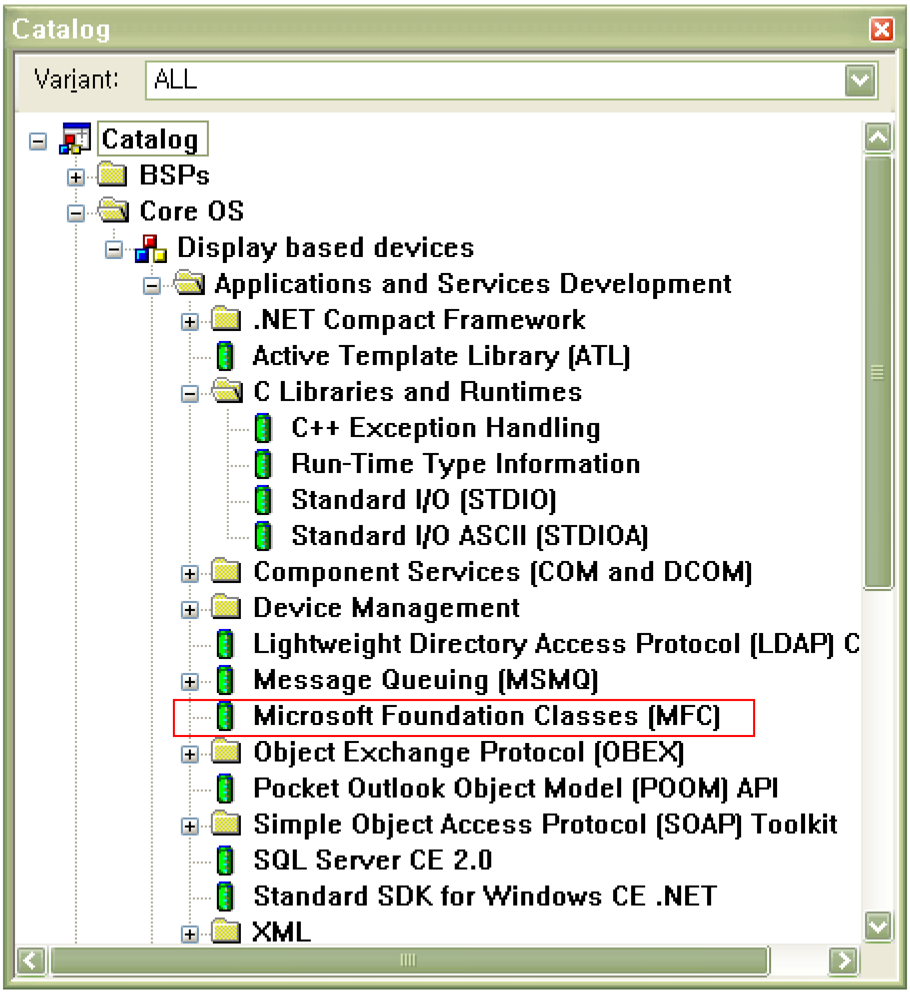


Now eMbedded C++ debugger or remote tools can be used. This is an example of running Remote Registry Editor.



26. Other components

26.1 To run some applications



To run some applications, like MESPlayer, you have to add MFC component into your Windows CE image.

27. Example of BSP Components Setting

This example shows components added to build sample image(BSP2.2.0) . This is not optimized for current settings and some unnecessary components might be included.

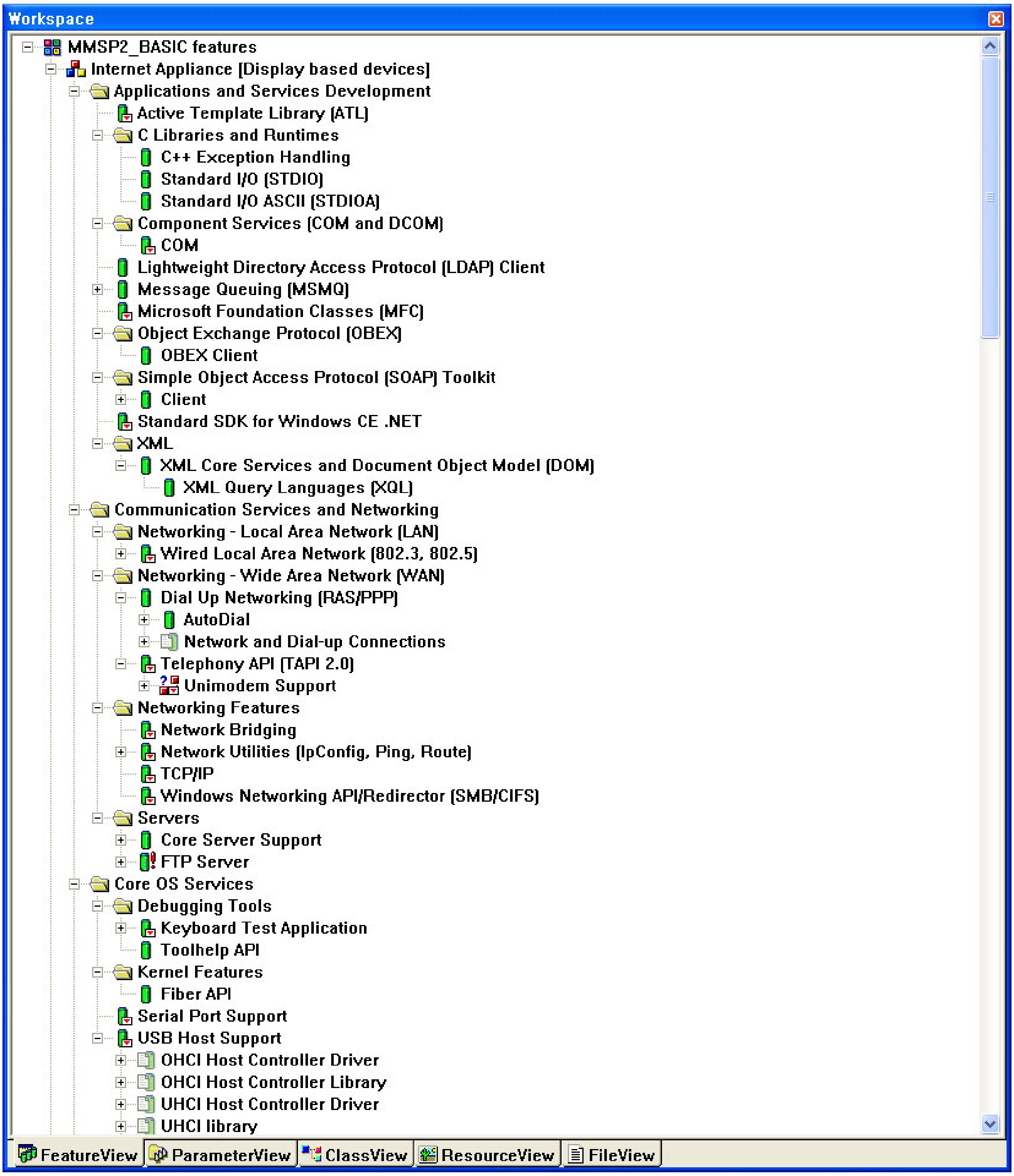


Figure 3-1. Example of BSP Components Setting 4-1

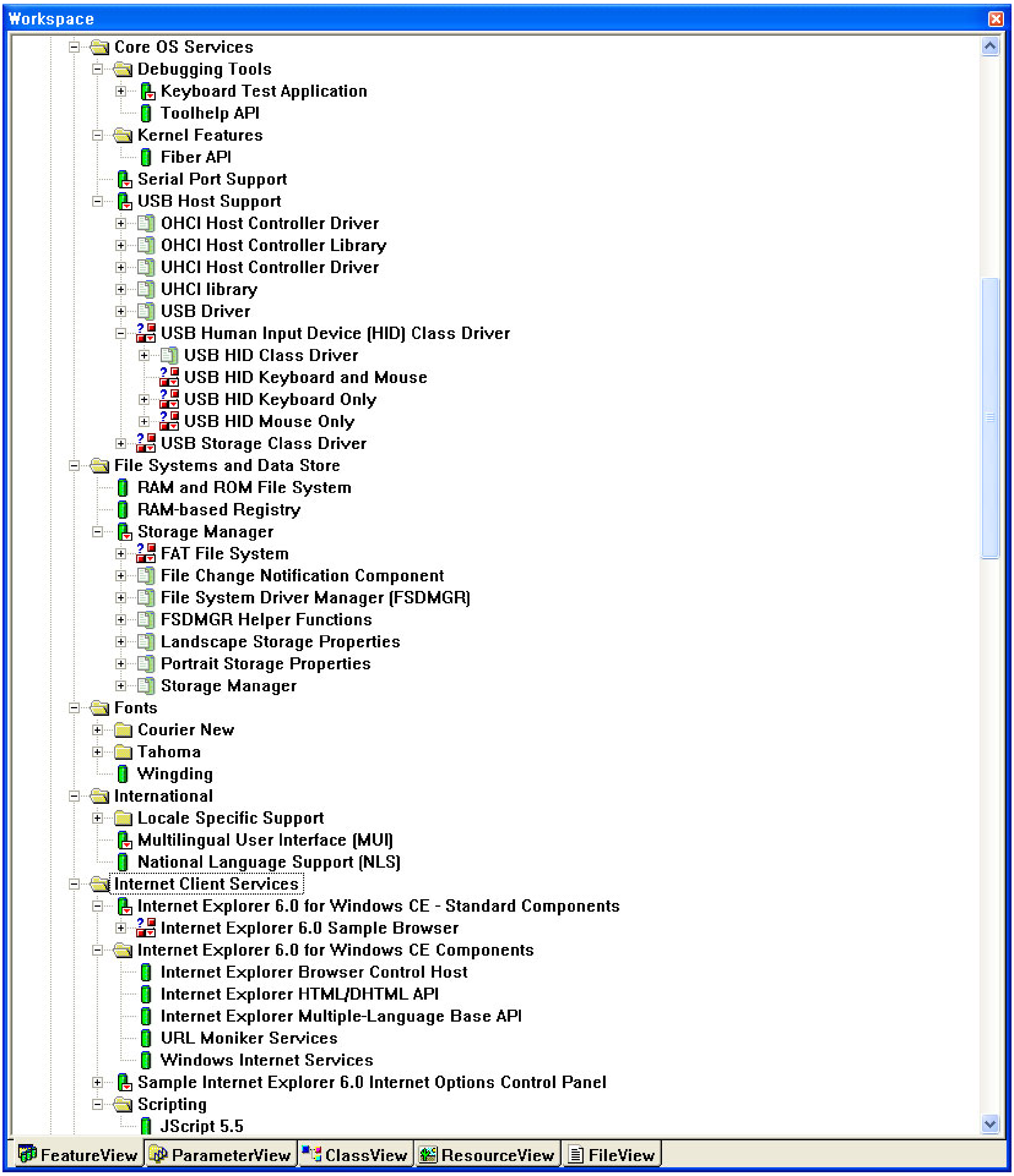


Figure 3-2. Example of BSP Components Setting 4-2

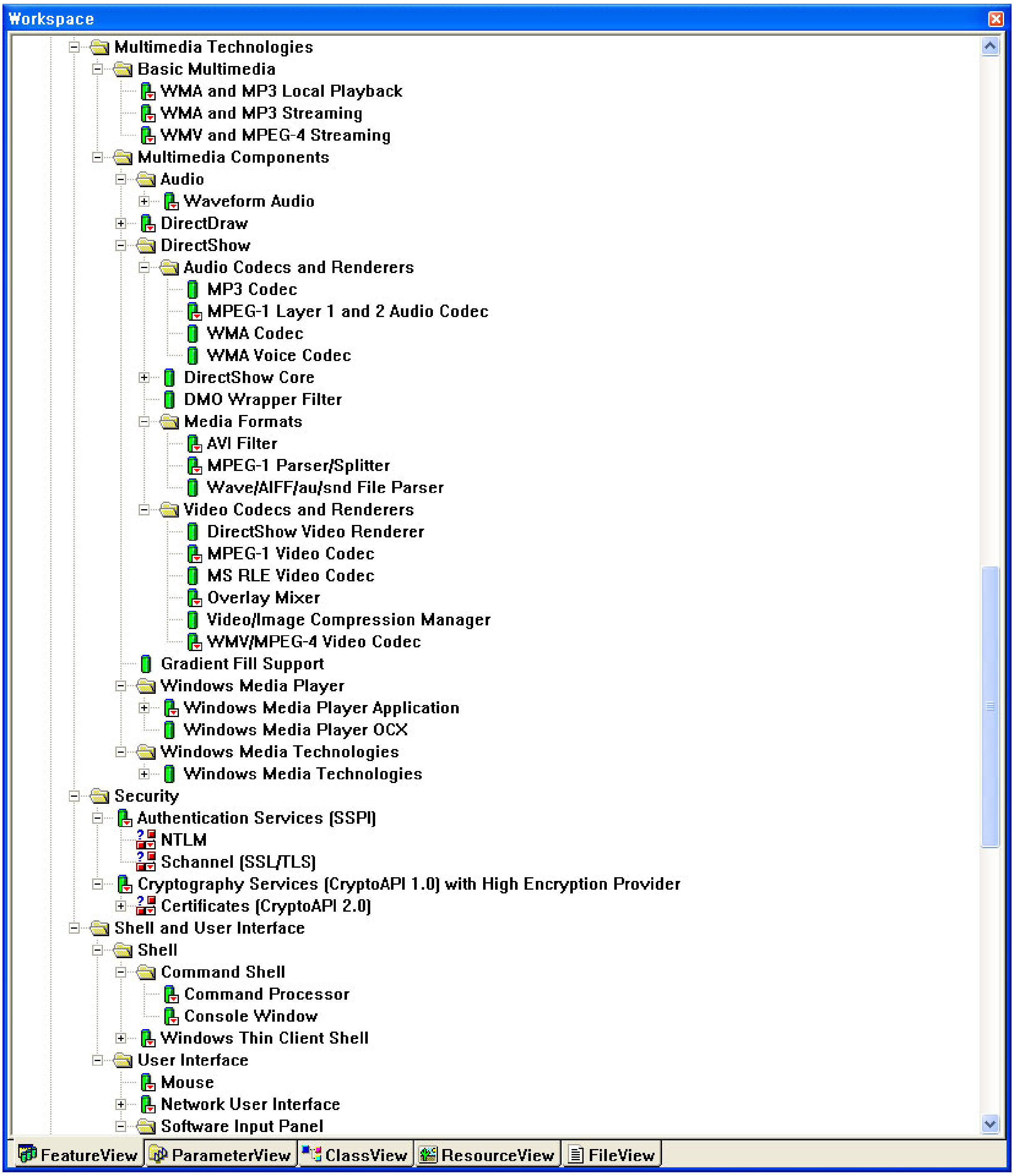


Figure 3-3. Example of BSP Components Setting 4-3

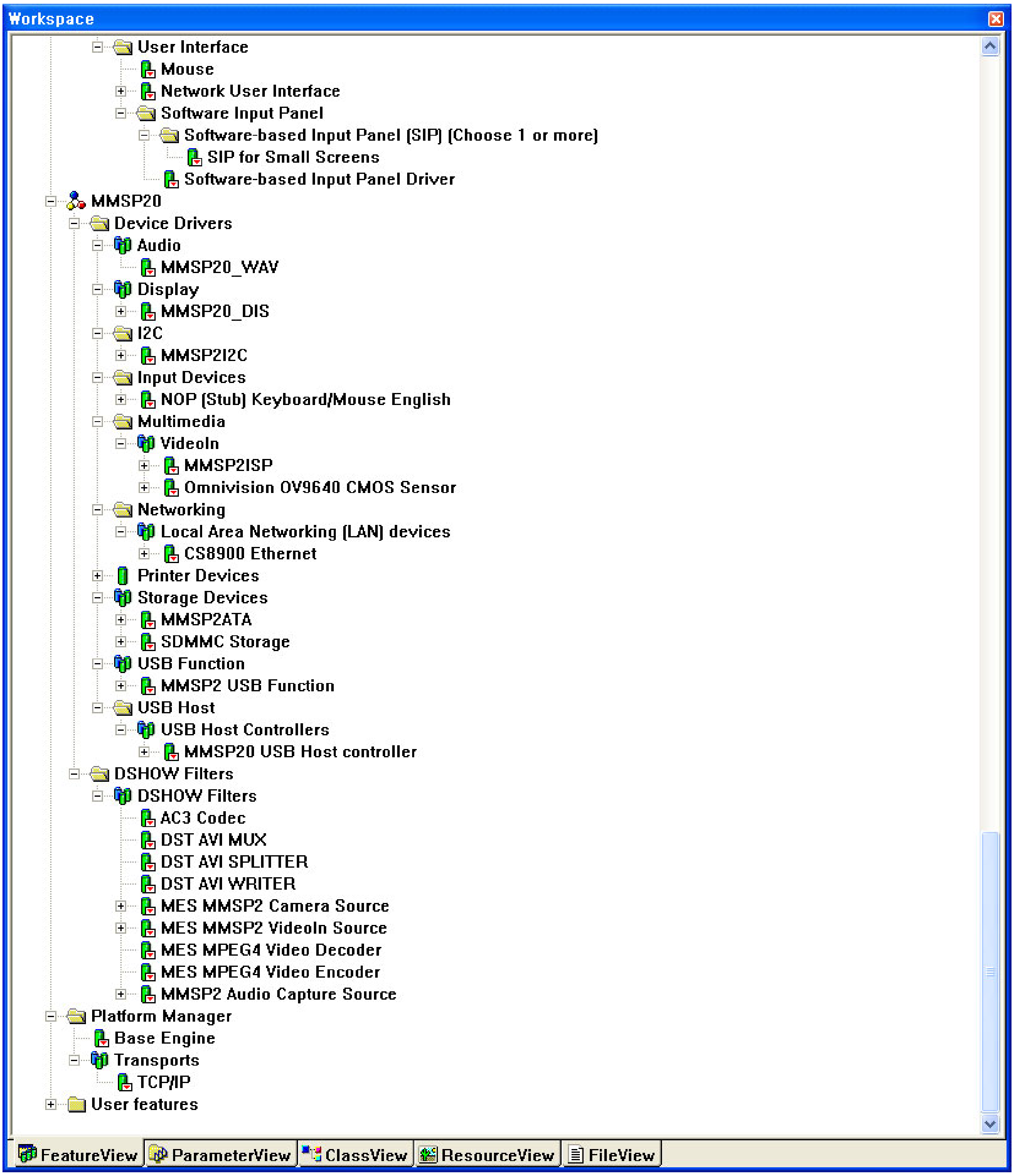


Figure 3-4. Example of BSP Components Setting 4-4